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ORIGINAL PAPER

Characterizing the U–Pb systematics of baddeleyite through chemical abrasion: application of multi-step digestion methods to baddeleyite geochronology

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Abstract U–Pb baddeleyite geochronology has become a major tool for dating mafic rocks, especially dikes associated with Large Igneous Provinces. However, in many cases, post-crystallization Pb-loss and intergrowth of baddeleyite and zircon limit the precision and/or accuracy of crystallization ages. We present results from multi-step digestion experiments designed to understand and reduce these effects. Experiments were carried out on Neoproterozoic baddeleyites with zircon inter- and over-growths from the Gannakouriep dike swarm, Namibia, and on fragments of a large Paleoproterozoic baddeleyite from the Phalaborwa carbonatite, South Africa. Multi-step digestion experiments on annealed Phalaborwa baddeleyite were designed to test whether the recently developed chemical abrasion technique for U–Pb zircon geochronology can be applied to baddeleyite. The experiments generated complex results—individual digestion steps were both normally and reversely discordant suggesting that U and Pb were decoupled in the multi-step digestions—and indicate that the current form of multi-step chemical abrasion is not an effective method for reducing the impact of Pb-loss in baddeleyite. A separate set of experiments on the Gannakouriep baddeleyite focused on isolating the zircon and baddeleyite components in composite grains. Conventional single-step digestion experiments for this sample resulted in a discordant suite of analyses with significant scatter attributed to inter- and over-grown zircon and highlight the difficulty of obtaining precise and accurate ages from composite grains. To isolate the baddeleyite and zircon in these grains, a two-step HCl–HF chemical abrasion procedure for annealed grains was developed. This technique was successful at selectively dissolving the baddeleyite and zircon components. Secondary zircon inter- and over-growths of baddeleyite can occur in samples affected by low-temperature alteration to granulite facies metamorphism, and the new HCl–HF chemical abrasion procedure provides a method for resolving both the igneous and metamorphic history of these composite grains.

Keywords Baddeleyite · Chemical abrasion · U/Pb · TIMS

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Introduction

Zircon ($ZrSiO_4$), baddeleyite (ZrO_2) and zirconolite ($CaZrTi_2O_7$) are the principal Zr-bearing phases in crustal rocks. Phase stability during igneous crystallization is controlled by a variety of factors, including the silica activity (Butterman and Foster 1967), oxygen fugacity (Carlier and Lorand 2008) and trace element content of the magma. Zircon is found in a wide range of igneous rocks with gabbroic to granitic compositions, whereas baddeleyite and zirconolite typically form in place of, or in addition to, zircon in intermediate to mafic rocks. All three minerals incorporate actinides (U, Th) but little or no Pb

during crystal growth (e.g., Heaman and LeCheminant 1993; Rasmussen and Fletcher 2004). In addition, diffusion experiments for zircon (Cherniak and Watson 2000) and inferences from geochronological studies of baddeleyite (Heaman and LeCheminant 2000) are consistent with slow diffusion rates for U and Pb, making these minerals excellent high-temperature chronometers.

Numerous studies have highlighted both the power and potential complexities of U–Pb baddeleyite geochronology. Baddeleyites have been used to date terrestrial and extra-terrestrial samples including lunar basalts (Rasmussen et al. 2008), Martian meteorites (Herd et al. 2007; Misawa and Yamaguchi 2007), carbonatites (Heaman and LeCheminant 1993), kimberlites-alnöites (Schärer et al. 1997; Heaman and LeCheminant 2000), anorthosites (Scoates and Chamberlain 1995), and numerous diabase dikes (e.g., LeCheminant and Heaman 1989; Hanson et al. 2004a, b; Söderlund et al. 2005) and gabbros (e.g., Davidson and van Breemen 1988; Heaman and Machado 1992; Heaman et al. 2002). Coupled baddeleyite geochronology and paleomagnetic analyses continue to play a particularly important role in indentifying, characterizing and reconstructing Large Igneous Provinces (LIP), providing constraints on supercontinent reconstructions and mantle dynamics. These applications require accurate, high-precision ages.

Baddeleyites can yield concordant high-precision dates, but they often show evidence for varying degrees of discordance (e.g., Heaman and Tarney 1989; Heaman and Machado 1992; Hanson et al. 2004a, b; Söderlund et al. 2005). Detailed studies of discordant baddeleyite populations suggest that baddeleyite can be affected by both primary discordance resulting from secular disequilibrium in the U–Pb decay chains or Pb-loss during radioactive decay, and secondary discordance resulting from later disturbance of the U–Pb system. Potential sources of primary discordance include Pb-loss by alpha recoil (Davis and Sutcliffe 1985), diffusional Rn loss (Heaman and LeCheminant 2000) and incorporation of excess Pa or exclusion of Th during baddeleyite crystallization (Amelin and Zaitsev 2002; Crowley and Schmitz 2009). These sources of primary discordance preferentially impact one of the two U-Pb decay chains (with the exception of alpha recoil), and typically result in relatively small percent discordance.

In this manuscript, we focus on the causes and impact of secondary discordance on U–Pb baddeleyite dating. Secondary discordance in baddeleyite has been attributed to two end-member models: post-crystallization Pb-loss and mixing of concordant baddeleyite with secondary zircon. Coeval and younger zircon inter- and over-growth of igneous baddeleyite can form during metamorphism or deuteric alteration and can be difficult to detect without detailed imaging (e.g., Davidson and van Breemen 1988; Heaman and LeCheminant 1993; Rasmussen et al. 2008). Secondary discordance can require the use of concordia upper intercept dates or restrict weighted mean calculations to a subset of the most concordant data, and thus limit the precision of baddeleyite dates for many samples. In addition, multi-component mixing between baddeleyite, secondary zircon and Pb-loss can lead to excessive scatter and inaccurate crystallization ages.

Recent developments in U–Pb zircon geochronology provide new means for minimizing or removing the effects of Pb-loss. Pb-loss in zircon is principally attributed to fast pathway diffusion related to radiation damage and defects in the crystal structure from U and Th decay (e.g., Silver and Deutsch 1963; Krogh and Davis 1974, 1975; Nasdala et al. 1998). Mattinson (2005) demonstrated that coupled annealing and multi-step digestion (chemical abrasion) can effectively remove damaged portions of zircon prior to analysis. In the chemical abrasion method, high-temperature annealing repairs radiation damage in zircon and prevents preferential leaching of Pb relative to U during multi-step digestions. Annealed zircons are dissolved in two or more steps: initial digestion steps preferentially dissolve the most damaged zircon—which is likely to be affected by post-crystallization Pb-loss—isolating the highest quality low-U zircon for final analysis. Coupled Raman spectroscopy, X-ray diffraction analyses and multi-step digestion experiments suggest that the differential solubility of zircon is related to residual radiation damage in the crystal lattice, which is not fully repaired during the high-temperature annealing (Mattinson et al. 2007). The chemical abrasion technique has proven highly effective at removing damaged lattice from both grain rims and cores—accessing material that is difficult to isolate using standard mechanical abrasion—and is now routinely applied to single and multi-grain zircon analyses (e.g., Mundil et al. 2004; Condon et al. 2005; Schoene et al. 2006; Crowley et al. 2007; Rioux et al. 2007).

The causes of Pb-loss in baddeleyite are not well understood. Zirconia (ZrO_2) has three stable polymorphs (Subbarao 1981): monoclinic baddeleyite ($\leq$1,170°C, 1 atm), tetragonal zirconia (1,170–2,370°C, 1 atm) and cubic zirconia ($\geq$2,370°C, 1 atm). Yttria-stabilized cubic zirconia has been proposed as both an inert matrix fuel for nuclear reactors (Oversby et al. 1997; Degueldre and Paratte 1998; Degueldre 2007) and a medium for radioactive waste storage (Weber et al. 1998; Ewing 1999), and extensive research indicates that it can withstand high doses of radiation (e.g. Sickafus et al. 1999a, b). Neutron and ion (Bi^{+}, Ne^{+}, Xe^{++}, Kr^{++}) irradiation experiments on monoclinic zirconia indicate that low to moderate radiation doses (0.1–30 displacements per atom) induce a phase transformation to cubic or tetragonal zirconia, but do not lead to amorphization of the crystal structure (e.g., Sickafus

et al. 1999a; Simeone et al. 2006; Valdez et al. 2008). The existing data on the effects of radiation damage from actinide decay in naturally occurring monoclinic baddeleyite are more limited, but a case study of baddeleyite with adjacent uranpyrochlore demonstrated that baddeleyite remained crystalline at alpha doses of 0.2–45 $\times$ 10^{15} mg^{-1} (Lumpkin 1999). These doses are well above the point of zircon metamictization (>8 $\times$ 10^{15} mg^{-1}; Murakami et al. 1991; Rios et al. 2000) and, together with the neutron and ion irradiation studies, suggest that amorphization is not an important factor in Pb-loss from baddeleyite. The high-alpha doses did lead to the development of small (5–20 nm) rotated domains (Lumpkin 1999) that could provide fast diffusion paths for Pb, however, it is not known whether these features exist in baddeleyite with more typical U and Th concentrations and alpha doses. Alternately, Heaman and LeCheminant (1993) suggested that discordant dates usually reflect mixing of baddeleyite and secondary zircon and that there is little or no diffusional Pb-loss in most baddeleyite.

In this contribution, we present experimental results on the application of chemical abrasion techniques to baddeleyite. The experiments were designed to understand and reduce the effects of Pb-loss and secondary zircon growth in discordant grains.

Sample description

For the digestion experiments, we focused on Neoproterozoic baddeleyites from the Gannakouriep dike swarm, Namibia, and Paleoproterozoic baddeleyite from the Phalaborwa carbonatite, South Africa. The Gannakouriep baddeleyites were separated from a felsic segregation at the centre of a ~1-km-wide medium-grained gabbro dike (05-48A). The sample has a hydrothermal overprint (biotite, amphibole and epidote) that is likely related to deuteric alteration. The analysed baddeleyite grains were typically ~100–400-μm-long thin-laths and plates. Electron backscatter images of a polished grain mount revealed that the baddeleyites are over-grown by thin zircon rims, and in many cases, zircon fills internal cracks within the grains (Fig. 1). Scanning electron images of individual baddeleyite grains show rough grain surfaces composed of fine-grained zircon over-growths (Fig. 2). The sample also contains limited 100–200-μm-long euhedral igneous zircon grains, which vary from clear-colourless to opaque and metamic. These grains were dated to provide an independent constraint on the crystallization age of the sample.

The Phalaborwa carbonatite contains millimetre- to centimetre-sized baddeleyite grains. Previous analyses of Phalaborwa baddeleyite yielded concordant to slightly discordant (<1%) U–Pb data with a weighted mean $^{207}Pb/^{206}Pb$ date of 2059.60 ± 0.35 Ma (n = 59; MSWD = 7.7; Heaman 2009). For the digestion experiments, we broke a single large baddeleyite into ~100–200-μm fragments for analysis (PhalaSG). Phalaborwa baddeleyite can contain apatite, carbonate, barite, celestite, feldspar, magnetite, monazite, uranothorianite and zircon inclusions (Heaman 2009). However, electron backscatter imaging of several grain fragments from the sample used in this study identified only carbonate inclusions, and care was taken to avoid inclusions in the analysed grains—there is no evidence of zircon inter- or over-growths on the analysed baddeleyite.

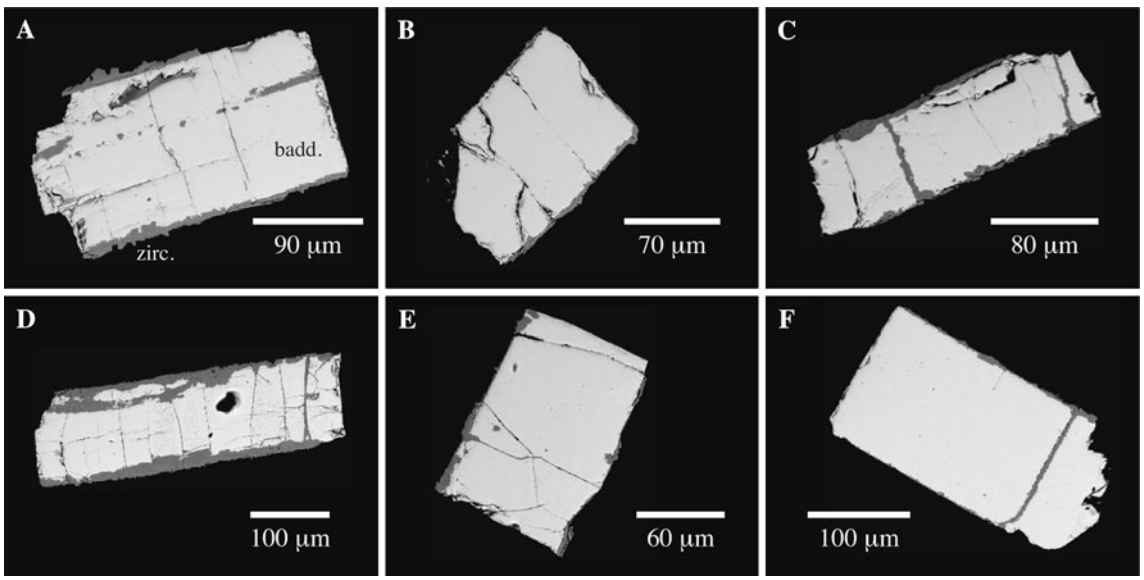


Fig. 1 Backscatter electron images of baddeleyite from sample 05-48A in a polished grain mount. The baddeleyite are over-grown by zircon rims (*darker grey*), and in some cases, zircon fill cracks within the grains

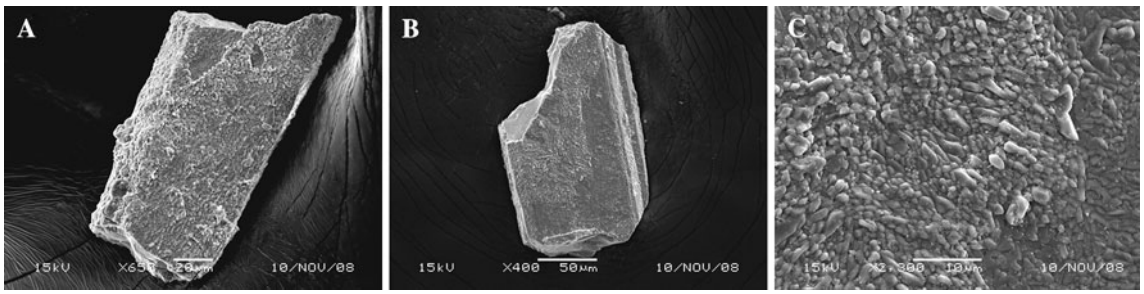


Fig. 2 **a**, **b** Scanning electron microscope images of baddeleyite over-grown by zircon rims from sample 05-48A. **c** Close up of the fine-grained zircon rim on the grain shown in **b**

To complement the digestion experiments, we analysed igneous zircon fragments and baddeleyite separated from a heavy mineral separate from the Phalaborwa mine (PhalaHMS).

Analytical methods

To develop a method to reduce the effects of Pb-loss and secondary zircon growth in baddeleyite geochronology, we experimented with a range of different digestion procedures using nitric (HNO_3), hydrochloric (HCl) and hydrofluoric (HF) acids. The digestion procedures and experimental results are described below. All experiments were carried out on single grains or grain fragments and were spiked with the EARTHTIME ^{205}Pb–^{233}U–^{235}U (ET535) tracer. Following digestion, leachates and sample solutions were converted to 3.1 N HCl, and U and Pb were separated on anion exchange columns prior to analysis (Krogh 1973). For experiments on annealed samples, baddeleyite grains were annealed in quartz crucibles at 900°C for 60 h.

Isotopic analyses were carried out on a VG sector 54 thermal ionization mass spectrometer. Samples were loaded in silica gel (Gerstenberger and Haase 1997) onto out-gassed Re filaments. Pb isotopes were measured by peak-hopping on the Daly detector and fractionation corrected based on repeat analyses of NBS 981. U isotopes were measured statically on Faraday cups and fractionation corrected using the ^{233}U–^{235}U double spike. All data reduction and error propagation was done using the U–Pb Redux software package (Bowring et al. 2008; McLean et al. 2008).

Lead isotopic ratios were corrected for common Pb based on measured $^{206}Pb/^{204}Pb$ and $^{207}Pb/^{204}Pb$. For baddeleyite analyses, common Pb was partitioned between laboratory blank and initial sample Pb. Initial common Pb values for 05-48A were estimated by stepwise dissolution of plagioclase following Housh and Bowring (1991; electronic supplementary material) and initial common Pb values for Phalaborwa were estimated from Stacey and Kramers (1975) to be consistent with Heaman (2009). For zircon analyses, all common Pb was assumed to be laboratory blank Pb isotopic ratios were determined by repeat analyses of spiked total procedural blanks.

Experimental procedures and results

HNO_3 rinse and single-step HF digestion of un-annealed baddeleyite (05-48A)

Initial single-step digestion experiments were designed to characterize the Gannakouriep baddeleyite and study the effects of annealing and pre-digestion cleaning procedures on U–Pb systematics. These experiments followed the traditional approach for baddeleyite geochronology (Krogh et al. 1987). Individual un-annealed baddeleyite grains were loaded into Teflon beakers and fluxed in ~7 N HNO_3 on a ~110°C hot plate. The baddeleyites were then rinsed in H_2O, loaded into Teflon microcapsules with ~120 µl of 29 N HF and ~10 µl of spike (1 N HNO_3) and dissolved in Parr acid digestion vessels held at 210°C for 48 h. To determine the impact of the HNO_3 flux, we broke several grains in half and subjected one half to a HNO_3 flux and one half to a H_2O flux on the same hot plate. We varied the length of the hot plate fluxes from 1 to 25 h and used both 7 N and 15 N HNO_3 (Table 1).

The data from these experiments are variably discordant (Fig. 3; Table 1). A linear regression of the full data set yields an upper intercept date of 793.2 ± 1.0 Ma, but has an unexpectedly high MSWD (3.1, $n = 28$, 1 outlier excluded; Wendt and Carl 1991). Separate regressions for the H_2O-and HNO_3-fluxed samples yielded upper intercept dates of 793.9 + 1.3/−1.3 Ma (MSWD = 2.9, 1 outlier excluded; Fig. 3a) and 791.7 + 1.7/−1.6 Ma (MSWD = 3.3; Fig. 3b), respectively. Comparison of data from the grain fragments that were subjected to separate HNO_3 and H_2O rinses shows some variability in the $^{206}Pb/^{238}U$ and $^{207}Pb/^{235}U$ of fragments from a single grain, but do not show a consistent sense of offset between the H_2O versus HNO_3 rinsing (Fig. 4).

Table 1 Single-step digestion experiments on 05-48A

	Rinse procedure[a]		Composition					Isotopic ratios					
	Reagent	*t* (h)	Pb*/Pbc[b]	Pb* (pg)[c]	Pbc (pg)[c]	U (pg)[c]	Th/U[d]	$^{206}Pb/^{204}Pb$[e]	$^{208}Pb/^{206}Pb$[f]	$^{206}Pb^*/^{238}U$[f]	2σ (% err)	$^{207}Pb^*/^{235}U$[f]	2σ (% err)
05-48A, non-annealed, H_2O rinse[j]													
b06	H_2O	1 h	42.4	185.80	4.38	1555.83	0.044	2852	0.01359	0.128770	0.166	1.1618	0.229
b07	H_2O	1 h	17.3	190.78	11.00	1491.06	0.340	1084	0.10465	0.127151	0.062	1.1457	0.193
b08	H_2O	1 h	64.7	311.26	4.81	2588.74	0.066	4306	0.02030	0.128841	0.070	1.1630	0.117
b09	H_2O	1 h	41.9	248.14	5.92	2123.19	0.058	2801	0.01786	0.125531	0.064	1.1314	0.125
b10a	H_2O	1 h	10.2	120.50	11.76	950.95	0.455	629	0.13995	0.122216	0.060	1.0992	0.296
b11a	H_2O	1 h	18.5	45.95	2.48	401.65	0.024	1262	0.00727	0.124155	0.087	1.1110	0.273
b12	H_2O	1 h	37.3	76.07	2.04	644.02	0.013	2532	0.00411	0.128491	0.057	1.1611	0.153
b13	H_2O	1 h	58.2	195.29	3.36	1632.24	0.041	3905	0.01261	0.129122	0.069	1.1663	0.125
b40a	H_2O	3 h	34.1	133.00	3.90	1155.18	0.248	2168	0.07639	0.117313	0.056	1.0510	0.154
b41a	H_2O	3 h	13.3	70.54	5.31	569.81	0.257	853	0.07897	0.125812	0.059	1.1335	0.252
b42a	H_2O	3 h	14.7	156.18	10.61	1296.74	0.422	903	0.13007	0.117180	0.055	1.0457	0.221
b43a	H_2O	3 h	22.1	74.77	3.38	599.30	0.450	1341	0.13858	0.120476	0.056	1.0813	0.186
b44a	H_2O	3 h	37.9	85.92	2.27	776.10	0.117	2496	0.03604	0.116994	0.063	1.0452	0.168
b54a	H_2O	25 h	13.2	46.16	3.49	416.91	0.421	815	0.13032	0.107754	0.070	0.9517	0.284
b55a	H_2O	25 h	16.3	36.14	2.21	324.27	0.090	1094	0.02767	0.118645	0.072	1.0649	0.276
05-48A, non-annealed, HNO_3 rinse[j]													
bb01	7 N HNO_3	1 h	22.1	265.06	12.00	2220.22	0.205	1426	0.06315	0.123074	0.053	1.1037	0.159
b02	7 N HNO_3	1 h	34.8	144.95	4.17	1175.05	0.372	2140	0.11462	0.121567	0.053	1.0915	0.150
b03	7 N HNO_3	1 h	21.9	197.56	9.00	1694.31	0.167	1433	0.05142	0.121466	0.056	1.0910	0.166
b04	7 N HNO_3	1 h	24.0	67.60	2.82	572.97	0.061	1614	0.01880	0.126599	0.068	1.1418	0.284
b05	7 N HNO_3	1 h	27.2	220.13	8.08	1899.44	0.043	1838	0.01323	0.125034	0.069	1.1244	0.157
b10b	7 N HNO_3	1 h	39.2	70.74	1.81	549.58	0.641	2252	0.19764	0.118460	0.055	1.0612	0.172
b11b	7 N HNO_3	1 h	12.3	22.52	1.83	196.94	0.036	842	0.01101	0.123648	0.093	1.1083	0.412
b40b	7 N HNO_3	3 h	88.6	177.99	2.01	1499.68	0.086	5865	0.02652	0.126460	0.065	1.1398	0.133
b41b	7 N HNO_3	3 h	51.3	150.88	2.94	1246.99	0.098	3387	0.03025	0.128472	0.083	1.1582	0.162
b42b	7 N HNO_3	3 h	48.3	185.78	3.84	1488.93	0.360	2975	0.11080	0.123370	0.082	1.1071	0.169
b43b	7 N HNO_3	3 h	37.6	82.84	2.20	674.31	0.424	2280	0.13053	0.119439	0.074	1.0723	0.218
b44b	7 N HNO_3	3 h	12.5	85.43	6.81	708.04	0.155	828	0.04772	0.126082	0.098	1.1368	0.314
b54b	15 N HNO_3	25 h	41.9	47.34	1.13	460.34	0.316	2620	0.09801	0.102917	0.078	0.9055	0.209
b55b	15 N HNO_3	25 h	22.5	16.57	0.74	142.37	0.110	1499	0.03375	0.123207	0.110	1.1112	0.467
05-48A, annealed, H_2O rinse[j]													
bb17a	H_2O	14 h	50.8	59.89	1.18	498.24	0.113	3350	0.03465	0.127112	0.063	1.1460	0.143
b18a	H_2O	14 h	25.5	54.84	2.15	471.57	0.014	1735	0.00438	0.126522	0.060	1.1381	0.197

Table 1 continued

	Rinse procedure[a]		Composition					Isotopic ratios					
	Reagent	t (h)	Pb*/Pbc[b]	Pb* (pg)[c]	Pbc (pg)[c]	U (pg)[c]	Th/U[d]	$^{206}Pb/^{204}Pb$[e]	$^{208}Pb/^{206}Pb$[f]	$^{206}Pb^*/^{238}U$[f]	2σ (% err)	$^{207}Pb^*/^{235}U$[f]	2σ (% err)
b19a	H_2O	24 h	17.9	146.03	8.15	1253.38	0.190	1165	0.05866	0.120586	0.061	1.0829	0.192
b20a	H_2O	24 h	12.1	35.13	2.91	331.72	0.077	817	0.02368	0.113221	0.076	1.0097	0.317
b21a	H_2O	38 h	15.7	73.44	4.69	606.54	0.103	1045	0.03159	0.128417	0.145	1.1574	0.240
b22a	H_2O	38 h	6.9	87.45	12.73	707.64	0.215	454	0.06608	0.127030	0.067	1.1443	0.384
b23a	H_2O	12 h	9.1	49.05	5.40	471.42	0.213	596	0.06576	0.107019	0.093	0.9596	0.372
b52a	H_2O	25 h	21.0	47.38	2.26	464.32	0.042	1419	0.01291	0.110147	0.089	0.9877	0.281
b53a	H_2O	25 h	37.2	50.47	1.36	445.96	0.021	2526	0.00642	0.122920	0.067	1.0994	0.181
05-48A, annealed, HNO_3 rinse[j]													
bb17b	7 N HNO_3	14 h	20.9	63.09	3.02	530.38	0.164	1364	0.05049	0.124010	0.062	1.1159	0.178
b18b	7 N HNO_3	14 h	22.1	97.13	4.39	825.98	0.017	1505	0.00527	0.127819	0.058	1.1504	0.165
b19b	7 N HNO_3	24 h	34.1	115.88	3.40	1013.77	0.151	2223	0.04657	0.119589	0.056	1.0759	0.134
b20b	7 N HNO_3	24 h	23.6	71.81	3.04	636.01	0.051	1590	0.01559	0.121555	0.059	1.0927	0.170
b21b	7 N HNO_3	38 h	49.2	97.82	1.99	808.21	0.095	3253	0.02907	0.128647	0.058	1.1608	0.135
b22b	7 N HNO_3	38 h	23.5	158.30	6.75	1299.38	0.128	1545	0.03946	0.128271	0.058	1.1561	0.159
b23b	15 N HNO_3	12 h	9.5	47.14	4.95	399.60	0.064	650	0.01979	0.126484	0.063	1.1402	0.305
b52b	15 N HNO_3	25 h	13.7	28.78	2.10	258.69	0.031	939	0.00960	0.120438	0.121	1.0839	0.440
b53b	15 N HNO_3	25 h	18.6	25.84	1.39	240.43	0.040	1262	0.01242	0.116148	0.088	1.0281	0.326
Phalaborwa, annealed, HNO_3 rinse[k]													
b5 I	7 N HNO_3	25 h	0.2	2.63	15.82	68.51	0.210	29	0.06633	0.037917	3.471	0.5920	25.637
b5 II	–	–	6197.8	28214.16	4.55	78046.29	0.010	402476	0.00289	0.372144	0.112	6.5267	0.135

	Isotopic ratios			Dates						
	$^{207}Pb^*/^{206}Pb^*$[f]	2 (% err)	corr.[g] coef.	$^{206}Pb/^{238}U$[h]	2σ (abs)	$^{207}Pb/^{235}U$[h]	2σ (abs)	$^{207}Pb/^{206}Pb$[h]	2σ (abs)	% disc[i]
05-48A, non-annealed, H_2O rinse[j]										
b06	0.065436	0.126	0.840	780.84	1.22	782.80	1.25	788.40	2.66	0.96
b07	0.065350	0.161	0.630	771.59	0.45	775.20	1.05	785.61	3.39	1.78
b08	0.065469	0.070	0.824	781.25	0.52	783.37	0.64	789.43	1.50	1.04
b09	0.065367	0.074	0.877	762.32	0.46	768.41	0.68	786.16	1.59	3.03
b10a	0.065227	0.264	0.595	743.30	0.42	752.94	1.57	781.68	5.56	4.91
b11a	0.064903	0.236	0.551	754.43	0.62	758.68	1.46	771.21	4.97	2.18
b12	0.065540	0.114	0.783	779.25	0.42	782.49	0.84	791.72	2.41	1.57
b13	0.065508	0.071	0.879	782.85	0.51	784.89	0.68	790.68	1.52	0.99
b40a	0.064975	0.124	0.658	715.08	0.38	729.37	0.80	773.53	2.62	7.56
b41a	0.065346	0.223	0.575	763.93	0.43	769.44	1.36	785.48	4.69	2.74

Table 1 continued

	Isotopic ratios			Dates						
	$^{207}Pb*/^{206}Pb*$[f]	2 (% err)	corr.[g] coef.	$^{206}Pb/^{238}U$[h]	2σ (abs)	$^{207}Pb/^{235}U$[h]	2σ (abs)	$^{207}Pb/^{206}Pb$[h]	2σ (abs)	% disc[i]
b42a	0.064720	0.193	0.592	714.31	0.37	726.74	1.15	765.26	4.08	6.66
b43a	0.065095	0.157	0.629	733.30	0.39	744.27	0.98	777.42	3.31	5.67
b44a	0.064796	0.128	0.734	713.24	0.43	726.52	0.87	767.70	2.71	7.09
b54a	0.064058	0.250	0.579	659.69	0.44	678.99	1.41	743.54	5.29	11.28
b55a	0.065094	0.240	0.585	722.76	0.49	736.22	1.44	777.37	5.07	7.02
05-48A, non-annealed, HNO_3 rinse[j]										
bb01	0.065039	0.124	0.744	748.23	0.37	755.12	0.84	775.59	2.62	3.53
b02	0.065118	0.118	0.692	739.58	0.37	749.22	0.79	778.14	2.51	4.96
b03	0.065142	0.131	0.707	739.00	0.39	748.98	0.88	778.94	2.78	5.13
b04	0.065413	0.247	0.625	768.43	0.49	773.37	1.54	787.64	5.19	2.44
b05	0.065219	0.116	0.726	759.47	0.49	765.06	0.84	781.40	2.46	2.81
b10b	0.064971	0.138	0.711	721.69	0.38	734.41	0.90	773.39	2.91	6.68
b11b	0.065006	0.365	0.590	751.53	0.66	757.34	2.20	774.54	7.68	2.97
b40b	0.065370	0.087	0.815	767.64	0.47	772.42	0.72	786.26	1.86	2.37
b41b	0.065382	0.112	0.755	779.14	0.61	781.09	0.88	786.66	2.38	0.96
b42b	0.065083	0.136	0.604	749.93	0.58	756.77	0.90	777.02	2.87	3.49
b43b	0.065113	0.185	0.576	727.33	0.51	739.86	1.14	777.98	3.90	6.51
b44b	0.065392	0.271	0.568	765.48	0.70	770.98	1.70	786.96	5.69	2.73
b54b	0.063809	0.167	0.659	631.48	0.47	654.64	1.01	735.30	3.55	14.12
b55b	0.065409	0.422	0.503	749.00	0.78	758.73	2.50	787.53	8.87	4.89
05-48A, annealed, H_2O rinse[j]										
bb17a	0.065385	0.089	0.907	771.37	0.45	775.33	0.78	786.76	1.89	1.96
b18a	0.065238	0.160	0.705	767.99	0.43	771.59	1.06	782.02	3.37	1.79
b19a	0.065132	0.151	0.748	733.93	0.42	745.05	1.01	778.59	3.19	5.74
b20a	0.064680	0.267	0.716	691.43	0.50	708.73	1.62	763.93	5.64	9.49
b21a	0.065366	0.170	0.711	778.83	1.06	780.72	1.31	786.14	3.58	0.93
b22a	0.065333	0.354	0.524	770.90	0.48	774.55	2.08	785.09	7.44	1.81
b23a	0.065030	0.339	0.463	655.41	0.58	683.07	1.85	775.31	7.13	15.47
b52a	0.065035	0.237	0.607	673.60	0.57	697.54	1.42	775.46	5.00	13.14
b53a	0.064870	0.144	0.667	747.35	0.47	753.07	0.96	770.11	3.06	2.96
05-48A, annealed, HNO_3 rinse[j]										
bb17b	0.065263	0.142	0.692	753.60	0.44	761.01	0.95	782.83	2.99	3.73
b18b	0.065275	0.126	0.769	775.41	0.43	777.43	0.90	783.23	2.65	1.00
b19b	0.065249	0.094	0.790	728.20	0.39	741.62	0.70	782.36	2.01	6.92

Table 1 continued

	Isotopic ratios			Dates						
	$^{207}Pb^*/^{206}Pb^*$[f]	2 (% err)	corr.[g] coef.	$^{206}Pb/^{238}U$[h]	2σ (abs)	$^{207}Pb/^{235}U$[h]	2σ (abs)	$^{207}Pb/^{206}Pb$[h]	2σ (abs)	% disc[i]
b20b	0.065195	0.129	0.777	739.51	0.41	749.80	0.90	780.63	2.73	5.27
b21b	0.065445	0.099	0.746	780.14	0.42	782.35	0.74	788.66	2.10	1.08
b22b	0.065369	0.120	0.762	777.99	0.42	780.12	0.86	786.22	2.54	1.05
b23b	0.065380	0.280	0.468	767.78	0.46	772.61	1.65	786.60	5.90	2.39
b52b	0.065273	0.404	0.423	733.08	0.84	745.54	2.33	783.14	8.49	6.39
b53b	0.064195	0.293	0.488	708.35	0.59	717.95	1.68	748.07	6.19	5.31
Phalaborwa, annealed, HNO_3 rinse[k]										
b5 I	0.113240	24.059	0.508	239.91	8.17	472.16	96.80	1852.06	434.94	87.05
b5 II	0.127199	0.044	0.949	2039.48	1.96	2049.51	1.19	2059.62	0.81	0.98

[a] Grains rinsed in H_2O or HNO_3 on ~110°C hot plate prior to digestion

[b] Ratio of radiogenic to common Pb

[c] Total radiogenic Pb, common Pb, and U (picograms)

[d] Th/U ratio calculated from $^{208}Pb/^{206}Pb$ and the $^{206}Pb/^{238}U$ date of the sample

[e] Fractionation and spike corrected $^{206}Pb/^{204}Pb$

[f] Fractionation, spike, and common Pb corrected radiogenic isotope ratios. Initial Pbc was corrected using $^{206}Pb/^{204}Pb = 17.89 \pm 0.10$, $^{207}Pb/^{204}Pb = 15.59 \pm 0.10$, and $^{208}Pb/^{204}Pb = 37.76 \pm 0.200$ (2σ absolute errors). Laboratory Pb blanks ranged from 0.56 to 2.25 pg and were corrected using $^{206}Pb/^{204}Pb = 18.24 \pm 0.42$, $^{207}Pb/^{204}Pb = 15.34 \pm 0.31$ and $^{208}Pb/^{204}Pb = 37.35 \pm 0.99$ (2σ absolute errors)

[g] Correlation coefficient of radiogenic $^{207}Pb^*/^{235}U$ and $^{206}Pb^*/^{238}U$

[h] Dates (Ma) calculated using $^{238}U/^{235}U = 137.88$, and decay constants of $^{238}U = 1.5513 \times 10^{-10}$ year^{-1} and $^{235}U = 9.8485 \times 10^{-10}$ year^{-1}. Uncertainties are 2σ absolute errors

[i] % discordance = 100 − (100 × ($^{206}Pb/^{238}U$ date)/($^{207}Pb/^{206}Pb$ date))

[j] Grain fractions labelled as "a" and "b" are fragments from a single grain that were subjected to different rinse treatments prior to digestion

[k] Phalaborwa b5 was a large fragment of baddeleyite subjected to a 25 h-7 N HNO_3 hot plate flux. I is the material removed in the flux and II is the composition of the residual solid

HNO_3 rinse and single-step HF digestion of annealed baddeleyite (05-48A and Phalaborwa)

The second group of experiments repeated the initial experimental procedures using annealed baddeleyite grains from the Gannakouriep sample. For all of these experiments, we broke the grains in half and subjected one half to a HNO_3 flux and the other half to a H_2O flux. We varied the length of the hot plate fluxes from 1 to 38 h and used both 7 N and 15 N HNO_3. In addition, we subjected a single large fragment of annealed Phalaborwa baddeleyite to a 24 h hot plate flux in 7 N HNO_3. This sample was spiked prior to fluxing and the resulting leachate was saved for analysis.

The results for the 05-48A analyses cover a similar range to the data from the un-annealed experiments (Fig. 3; Table 1). A linear regression of the data yields an upper intercept date of 787.4 + 1.3/−1.2 Ma (MSWD = 3.0, $n = 16$, 2 outliers excluded), which is distinctly younger than the upper intercept date from the un-annealed analyses. Separate regressions of the H_2O- and HNO_3-fluxed grains yielded upper intercept dates of 787.0 + 1.9/−1.8 Ma (MSWD = 4.5, 1 outlier excluded; Fig. 3c) and 787.1 + 1.8/−1.7 Ma (MSWD = 2.1, 1 outlier excluded; Fig. 3d), respectively. The similarity in the upper intercept dates is expected because all of the analyses in this experiment were done on broken grain pairs. As with the un-annealed samples, there are not systematic offsets between the data from HNO_3 and H_2O rinsed fragments (Fig. 4).

Combining the results from the annealed and un-annealed single-step digestion experiments yields an upper intercept date of 791.65 + 0.81/−0.79 Ma (MSWD = 6.7, $n = 44$, 3 excluded). The high MSWD of both the un-annealed (MSWD = 3.1), annealed (MSWD = 3.0) and combined (MSWD = 6.7) data sets indicates that the scatter in the data cannot be explained by our calculated analytical uncertainties and/or simple, two-component mixing. As a result, the calculated upper intercept dates are not statistically significant and the associated uncertainties do not reflect the true variation in the data.

The Phalaborwa experiment was designed to test whether baddeleyite is dissolved during hot plate fluxing in HNO_3 and to determine the isotopic composition of the dissolved material if present. The leachate from the hot plate flux contained a mixture of common Pb and radiogenic Pb (Pb*/

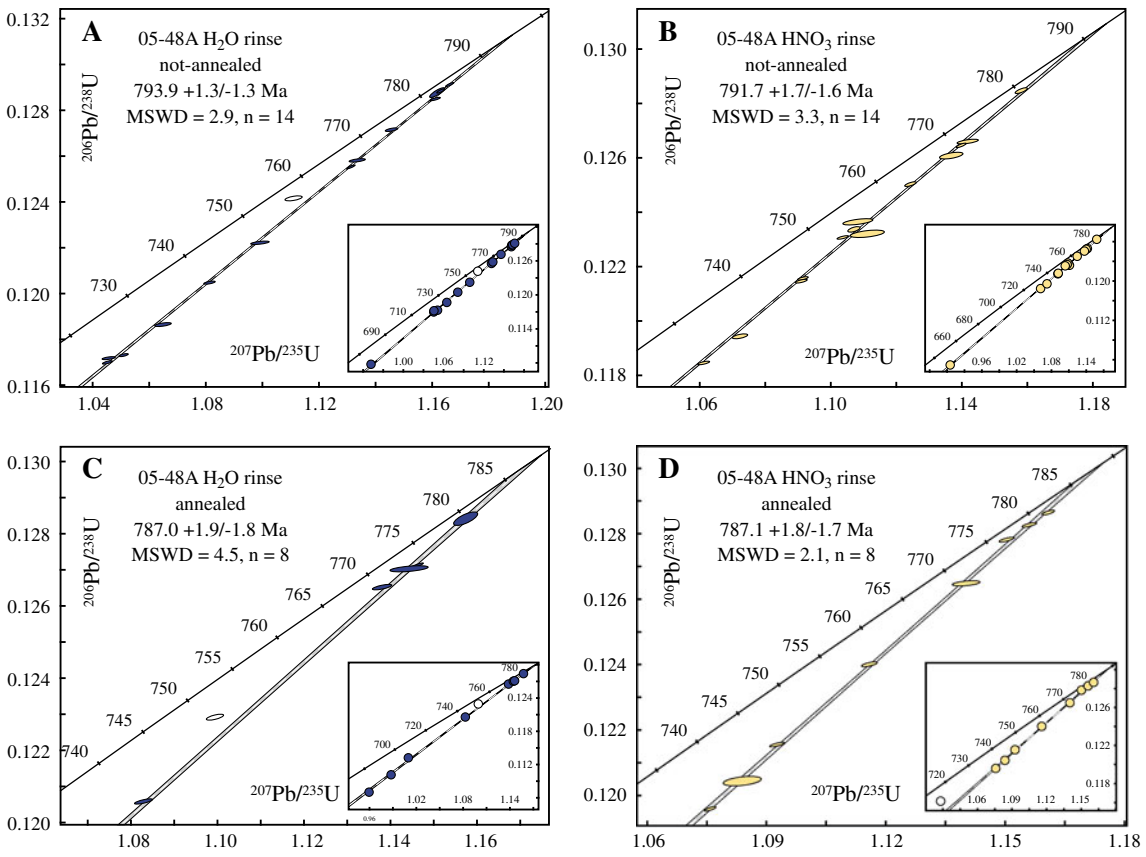


Fig. 3 U–Pb concordia diagrams for single-step baddeleyite data. *Inset* plots show the full data sets. Open data points were excluded from regressions. The *grey bands* are 2-sigma errors on the regression lines. **a** Un-annealed baddeleyite fluxed in H_2O. **b** Un-annealed baddeleyite fluxed in HNO_3. **c** Annealed baddeleyite fluxed in H_2O. **d** Annealed baddeleyite fluxed in HNO_3. Ages along concordia are in Ma. All error ellipses are 2-sigma. Plots and regressions were generated in U–Pb Redux (Bowring et al. 2008; McLean et al. 2008)

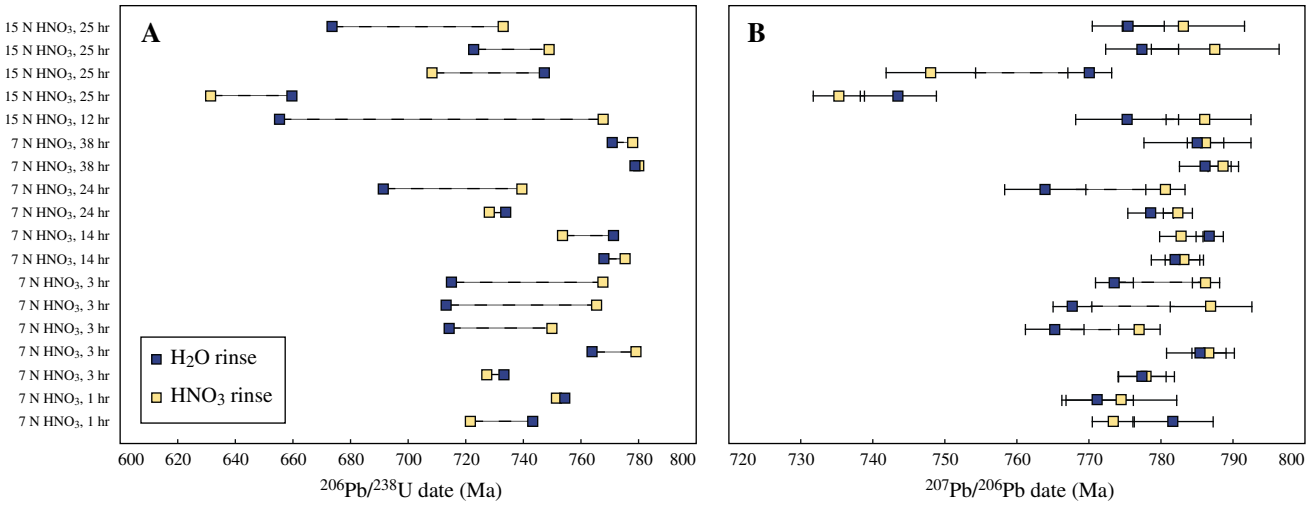


Fig. 4 Comparison of HNO_3- and H_2O-rinsed grain fragments from the single-step digestion experiments. Individual grains were broken in half and subjected to separate pre-treatments. The HNO_3 rinse conditions are listed along the *y*-axis. All fluxing was carried out on a 110°C hot plate. **a** Comparison of the $^{206}Pb/^{238}U$ dates of H_2O- and HNO_3-rinsed grain fragments. **b** Comparison of the $^{207}Pb/^{206}Pb$ dates of the H_2O- and HNO_3-rinsed grain fragments

Pbc = 0.17) and was highly discordant. The HF digestion of the main portion of the grain was 0.98% discordant and yielded a $^{207}Pb/^{206}Pb$ date of 2059.62 ± 0.81 Ma, which is consistent with prior analyses of Phalaborwa baddeleyites (Heaman 2009) and our new baddeleyite data from the PhalaHMS separate (discussed below). Mass balance calculations indicate that the leach step removed only 0.009% of the grain's total radiogenic Pb.

Multi-step HCl and HNO_3 digestion of annealed baddeleyite (PhalaSG)

Multi-step digestion experiments were carried out on three grain fragments of Phalaborwa baddeleyite (PhalaSG) to determine whether the chemical abrasion technique developed for zircon U–Pb dating can be used to minimize the impact of Pb-loss in baddeleyite geochronology. Annealed grains were subjected to a series of 6.2 N HCl (B4 and B6) or 7 N HNO_3 (B8) digestion steps followed by a final 29 N HF digestion (Table 2). HCl and HNO_3 were chosen for the multi-step digestions after initial experiments demonstrated that baddeleyite are soluble in these acids (discussed in more detail below). All steps were carried out in microcapsules within Parr acid digestion vessels. Digestion temperatures were increased sequentially from 125/160°C to 210°C, and each step lasted from 14 to 20 h. Following each digestion step, the grains were transferred into Teflon beakers and rinsed and sonicated in H_2O or HNO_3. Each step was spiked prior to digestion and the leachates were saved and analysed. Final HF digestions were held at 210°C for 48 h. An attempt at a multi-step experiment using 29 N HF digestion steps failed because the grain dissolved during the first 16 h, 120°C step.

The multi-step digestions generated complex results. The data for the three experiments plot along discordia lines with indistinguishable upper intercept dates of 2059.1 + 1.4/−1.0 Ma (B4; MSWD = 0.23), 2060.15 + 1.10/−0.99 Ma (B6; MSWD = 0.97) and 2060.33 + 0.65/−0.63 Ma (B8; MSWD = 0.71) and near zero age lower intercepts. Figure 5 shows the isotopic composition of the digestion steps and the evolution of the solid compositions, based on mass balance calculations. The first solid composition (a) corresponds to the initial grain compositions prior to any digestion, and ellipse A represents the material removed in the first digestion step. The inset plots in the right hand column show the cumulative percentage of Pb* released in the progressive digestion steps. For both HCl digestion experiments, the initial digestion steps were the most discordant and subsequent digestion steps had progressively higher $^{206}Pb/^{238}U$ and $^{207}Pb/^{235}U$. The two highest temperature HCl digestions steps in each experiment were reversely discordant (B4 C–D, 200–210°C and B6 H–I, 210°C). The final HF digestion steps (residue) for the experiments were more concordant than the initial bulk composition of the grain, although the HCl digestion steps for grain B6 removed a relatively small percentage of the total U and Pb* and there is only a small offset between the composition of the initial grain and the final residue.

The ten-step HNO_3 digestion experiment followed the same procedure as the grain B6 HCl experiment. In contrast to the HCl experiments, all of the HNO_3 digestion

Table 2 Multi-step chemical abrasion experiments on PhalaSG

	Digestion conditions[a]			Composition					Isotopic ratios					
	Reag.	T (°C)	t (h)	Pb*/Pbc[b]	Pb* (pg)[c]	Pbc (pg)[c]	U (pg)[c]	Th/U[d]	$^{206}Pb/^{204}Pb$[e]	$^{208}Pb/^{206}Pb$[f]	$^{206}Pb^*/^{238}U$[f]	2σ (% err)	$^{207}Pb^*/^{235}U$[f]	2σ (% err)
PhalaSG b4, multi-step 6.2 N HCl digestion														
b4a	HCl	160	14	42.5	111.08	2.61	307.20	0.019	2518	0.00535	0.371467	0.084	6.5091	0.165
b4b	HCl	180	14	325.9	295.62	0.91	812.44	0.008	19901	0.00219	0.374815	0.059	6.5718	0.095
b4c	HCl	200	14	0.3	1.51	5.09	2.71	0.852	30	0.23788	0.473116	6.135	8.4356	16.825
b4d	HCl	210	20	14.1	12.77	0.91	34.67	0.016	873	0.00448	0.378796	0.338	6.6386	0.435
b4e	HF	210	48	232.7	214.80	0.92	589.67	0.007	14201	0.00214	0.375264	0.061	6.5785	0.107
PhalaSG b6, multi-step 6.2 N HCl digestion														
b6a	HCl	125	16	0.2	0.24	1.30	9.86	0.938	26	0.29880	0.019641	16.255	0.3555	22.335
b6b	HCl	165	16	0.4	2.26	6.19	8.44	0.434	35	0.12848	0.249604	7.946	4.1389	15.560
b6c	HCl	170	16	2.0	1.94	0.97	6.09	0.024	141	0.00691	0.328472	3.300	5.6376	3.659
b6d	HCl	180	16	4.0	2.00	0.50	5.91	0.029	274	0.00824	0.347505	1.718	6.0355	1.991
b6e	HCl	190	16	7.3	3.91	0.54	11.31	0.029	483	0.00834	0.354176	1.247	6.2125	1.365
b6f	HCl	200	16	12.9	34.52	2.67	95.19	0.027	781	0.00762	0.371810	0.502	6.5132	0.594
b6g	HCl	210	16	70.0	102.53	1.46	282.74	0.014	4263	0.00393	0.372961	0.077	6.5410	0.116
b6h	HCl	210	16	5.6	8.33	1.50	22.47	0.033	352	0.00950	0.379550	0.482	6.6539	0.696
b6i	HCl	210	16	1.8	9.75	5.48	25.03	0.118	117	0.03376	0.389932	0.533	6.8522	2.159
B6j	HF	210	48	2841.5	3467.12	1.22	9494.02	0.011	174174	0.00307	0.375871	0.064	6.5926	0.105
PhalaSG b8, multi-step 7 N HNO_3 digestion														
b8a	HNO_3	125	16	0.3	0.50	1.78	9.86	0.723	31	0.22821	0.044515	3.250	0.6554	15.735
b8b	HNO_3	165	16	1.9	5.57	2.97	17.23	0.123	124	0.03570	0.323226	0.427	5.6721	1.894
b8c	HNO_3	170	16	26.0	14.76	0.57	42.33	0.017	1676	0.00502	0.358338	0.186	6.2802	0.258
b8d	HNO_3	180	16	9.2	7.91	0.86	22.61	0.013	596	0.00362	0.360012	0.363	6.3095	0.535
b8e	HNO_3	190	16	14.8	11.24	0.76	31.25	0.012	964	0.00351	0.369949	0.276	6.4936	0.411
b8f	HNO_3	200	16	56.9	44.29	0.78	122.35	0.010	3636	0.00295	0.372598	0.124	6.5356	0.193
b8g	HNO_3	210	16	122.8	111.27	0.91	306.32	0.009	7720	0.00271	0.374016	0.074	6.5583	0.119
b8h	HNO_3	210	16	3.1	2.22	0.71	6.14	0.019	217	0.00539	0.370545	1.225	6.5512	1.732
b8i	HNO_3	210	16	19.1	11.50	0.60	31.85	0.006	1243	0.00183	0.372222	0.644	6.5257	0.687
b8j	HF	210	48	2322.4	1237.26	0.53	3398.09	0.010	148645	0.00279	0.374828	0.059	6.5772	0.093

	Isotopic ratios			Dates						
	$^{207}Pb^*/^{206}Pb^*$[f]	2σ (% err)	corr.[g] coef.	$^{206}Pb/^{238}U$[h]	2σ (abs)	$^{207}Pb/^{235}U$[h]	2σ (abs)	$^{207}Pb/^{206}Pb$[h]	2σ (abs)	% disc[i]
PhalaSG b4, multi-step 6.2 N HCl digestion										
b4a	0.127087	0.110	0.798	2036.30	1.46	2047.13	1.45	2058.07	1.95	1.06
b4b	0.127164	0.053	0.840	2052.02	1.04	2055.57	0.83	2059.13	0.98	0.35
b4c	0.129315	14.807	0.491	2497.21	127.02	2279.02	152.73	2088.67	260.38	−19.56
b4d	0.127106	0.257	0.806	2070.66	5.99	2064.48	3.84	2058.33	4.55	−0.60

Table 2 continued

	Isotopic ratios			Dates						
	$^{207}Pb^*/^{206}Pb^*$[f]	2σ (% err)	corr.[g] coef.	$^{206}Pb/^{238}U$[h]	2σ (abs)	$^{207}Pb/^{235}U$[h]	2σ (abs)	$^{207}Pb/^{206}Pb$[h]	2σ (abs)	% disc[i]
b4e	0.127141	0.070	0.775	2054.12	1.06	2056.47	0.94	2058.82	1.26	0.23
PhalaSG b6, multi-step 6.2 N HCl digestion										
b6a	0.131275	21.575	0.410	125.39	20.19	308.86	59.48	2115.09	378.25	94.07
b6b	0.120263	12.514	0.601	1436.43	102.32	1662.02	127.25	1960.13	223.34	26.72
b6c	0.124477	2.101	0.823	1830.97	52.60	1921.86	31.56	2021.38	37.23	9.42
b6d	0.125966	0.908	0.890	1922.67	28.56	1980.99	17.34	2042.42	16.05	5.86
b6e	0.127218	0.496	0.932	1954.51	21.02	2006.21	11.94	2059.88	8.76	5.12
b6f	0.127048	0.334	0.827	2037.91	8.76	2047.68	5.22	2057.53	5.89	0.95
b6g	0.127198	0.068	0.816	2043.32	1.34	2051.44	1.02	2059.60	1.23	0.79
b6h	0.127146	0.578	0.570	2074.18	8.55	2066.52	6.14	2058.89	10.20	−0.74
b6i	0.127450	2.008	0.397	2122.51	9.64	2092.49	19.13	2063.09	35.41	−2.88
b6j	0.127209	0.062	0.825	2056.97	1.12	2058.36	0.93	2059.76	1.12	0.14
PhalaSG b8, multi-step 7 N HNO_3 digestion										
b8a	0.106783	14.683	0.416	280.76	8.93	511.80	63.26	1745.22	268.95	83.91
b8b	0.127273	1.756	0.424	1805.47	6.72	1927.13	16.35	2060.64	30.97	12.38
b8c	0.127109	0.153	0.809	1974.29	3.17	2015.69	2.26	2058.37	2.71	4.08
b8d	0.127110	0.374	0.715	1982.23	6.19	2019.78	4.69	2058.38	6.61	3.70
b8e	0.127305	0.293	0.700	2029.16	4.80	2045.04	3.62	2061.09	5.18	1.55
b8f	0.127217	0.118	0.802	2041.61	2.16	2050.71	1.70	2059.86	2.10	0.89
b8g	0.127174	0.065	0.862	2048.27	1.29	2053.76	1.05	2059.27	1.18	0.53
b8h	0.128227	1.086	0.783	2031.96	21.36	2052.81	15.26	2073.81	19.12	2.02
b8i	0.127153	0.211	0.951	2039.85	11.27	2049.38	6.05	2058.98	3.73	0.93
b8j	0.127264	0.042	0.921	2052.08	1.04	2056.29	0.82	2060.52	0.79	0.41

[a] All digestions carried out in Parr acid digestion vessels

[b] Ratio of radiogenic to common Pb

[c] Total radiogenic Pb, common Pb, and U (picograms)

[d] Th/U ratio calculated from $^{208}Pb/^{206}Pb$ and the $^{206}Pb/^{238}U$ date of the sample

[e] Fractionation and spike corrected $^{206}Pb/^{204}Pb$

[f] Fractionation, spike, and common Pb corrected radiogenic isotope ratios. Initial Pbc was corrected using $^{206}Pb/^{204}Pb = 15.04 \pm 0.30$, $^{207}Pb/^{204}Pb = 15.16 \pm 0.30$ and $^{208}Pb/^{204}Pb = 34.67 \pm 0.69$ (2σ absolute errors). Laboratory Pb blanks ranged from 0.47 to 0.73 pg and were corrected using $^{206}Pb/^{204}Pb = 18.24 \pm 0.42$, $^{207}Pb/^{204}Pb = 15.34 \pm 0.31$ and $^{208}Pb/^{204}Pb = 37.35 \pm 0.99$ (2σ absolute errors)

[g] Correlation coefficient of radiogenic $^{207}Pb^*/^{235}U$ and $^{206}Pb^*/^{238}U$

[h] Dates (Ma) calculated using $^{238}U/^{235}U = 137.88$, and decay constants of $^{238}U = 1.5513 \times 10^{-10}$ year^{-1} and $^{235}U = 9.8485 \times 10^{-10}$ year^{-1}. Uncertainties are 2σ absolute errors

[i] % discordance = $100 - (100 \times (^{206}Pb/^{238}U \text{ date})/(^{207}Pb/^{206}Pb \text{ date}))$

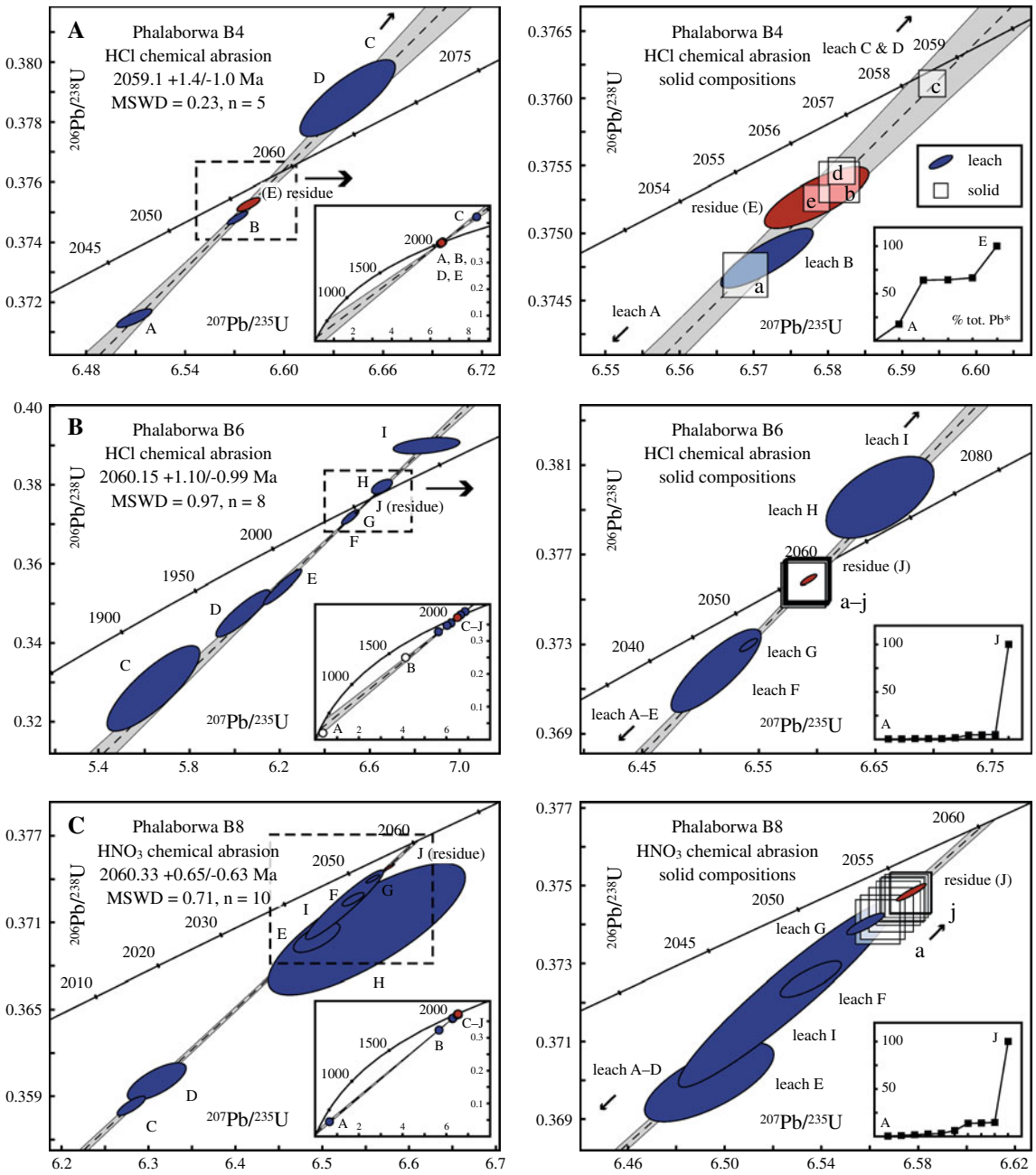


Fig. 5 U–Pb concordia plots of multi-step baddeleyite chemical abrasion data. The *left column* shows the composition of individual digestions steps (*ellipses*, *upper case letters*). The *inset* plots show the full data sets, and the *dashed box* shows the location of the enlarged plots in the right hand column. The *right column* shows the evolution of the solid composition (*white square*, *lower case letters*). The size of the solid square is proportional to the per cent of the solid remaining in each step. All solid compositions were calculated by mass balance. The *inset* plots in the right column show the cumulative percentage of Pb* released in the progressive digestion steps. The *ellipse colours* correspond to the reagent used in the digestion step (*red*, HF; *Blue*, HCl or HNO_3). Open data points were excluded from the regressions. Ages along concordia are in Ma. All *error ellipses* are 2-sigma. Plots and regressions were generated in U–Pb Redux (Bowring et al. 2008; McLean et al. 2008)

steps were more discordant than the initial bulk composition of the grain. The $^{206}Pb/^{238}U$ and $^{207}Pb/^{235}U$ of the digestion steps increase progressively for steps A–G (125–210°C) and then decrease in steps H and I. The final residue digestion was again the most concordant analysis.

Discrete HCl and HF digestion of annealed baddeleyite (05-48A)

Discrete digestion chemical abrasion experiments focused on isolating the baddeleyite and zircon components of

composite grains from the Gannakouriep sample. All grains were annealed prior to digestion to minimize the leaching of Pb* from the zircon rims during the two-step digestion (Mattinson 2005). Annealed grains were loaded into microcapsules with 6.2 N HCl, spiked and placed in Parr acid digestion vessels in a 210°C oven for 48 h. Following the HCl digestion, the leachates were pipetted out of the microcapsules and saved. The residual grains were rinsed repeatedly with H_2O, fluxed on a ~110°C hotplate in 7 N HNO_3 or 6.2 N HCl for >1 h and re-rinsed in H_2O. All rinsing and fluxing was done in the microcapsules to avoid losing grain fragments when transferring between microcapsules and beakers. The residual grains were then spiked and returned to the oven in ~29 N HF. In the two-step analyses, the final HF digestions were held at 210°C for 48 h. In the three-step digestions, we used intermediate low-temperature HF digestion steps to remove damaged zircon prior to final analyses. The low-temperature HF clean-up steps were held at 120–160°C for 6–8 h (digestion conditions are listed in Table 3). The residual grains were then rinsed and spiked, and final digestions were achieved in 29 N HF held at 210°C for 48 h.

The results of these experiments, the multi-step digestion experiments (described above) and single-step digestion experiments on both large fragments of Phalaborwa baddeleyite and single baddeleyite grains from the FC-4 Duluth anorthosite, all demonstrated that baddeleyite is soluble in HCl (some samples required >48 h digestion times). In the two-step HCl–HF experiments, the baddeleyite completely dissolved in the initial high-temperature HCl digestions steps, in some cases leaving behind hollow zircon shells (Fig. 6), which were dissolved in the subsequent HF digestion steps. These analyses document clear differences in dates between the baddeleyite grains and the inter- and over-grown zircon (Table 3). The analyses from the HCl (baddeleyite) digestion steps plot along a statistically significant discordia with an upper intercept date of 793.3 + 1.7/−1.5 Ma (MSWD = 0.64, $n = 13$; 0.45–9.75% discordant; Fig. 7a)—a single highly discordant grain (b49) with a distinct model Th/U ratio was excluded from the plot and regression. In contrast, analyses of all of the final HF digestion steps plot to the left of the HCl array (Fig. 7b, c). There is significant scatter in the HF digestion steps suggesting that they reflect multiple periods of zircon growth and/or Pb-loss. All of the analyses have uniformly lower $^{207}Pb/^{206}Pb$ dates than the baddeleyite digestion steps, and a linear regression of eight analyses that were pre-treated with intermediate low-temperature HF clean-up steps yield an upper intercept date of 767 + 11/−9.1 Ma (MSWD = 11), consistent with younger ages for the zircon rims but with excess scatter. Alternately, the data could also be explained by coeval crystallization of the baddeleyite and zircon followed by multiple periods of Pb-loss from the zircon rims.

Analyses of igneous zircon

Igneous zircons from both 05-48A and the Phalaborwa carbonatite were analysed to provide independent constraints on the crystallization ages of these samples (Fig. 8; Table 4). Zircon grains were dissolved following the chemical abrasion method (Mattinson 2005) using the analytical techniques for single grain analyses described by Schoene et al. (2006; leach conditions are listed in Table 4). Seven single grain analyses from sample 05-48A yielded concordant to slightly discordant (0.57%) data with a weighted mean $^{207}Pb/^{206}Pb$ date of 795.3 ± 0.6 Ma (MSWD = 0.15; Fig. 8a). Eleven analyses of Phalaborwa zircon yielded concordant to slightly discordant (0.39%) data with a weighted mean $^{207}Pb/^{206}Pb$ date of 2058.38 ± 0.41 Ma (MSWD = 0.80; Fig. 8b). Four baddeleyite grains from the same PhalaHMS separate yielded slightly discordant (0.31–0.56%) data with a weighted mean $^{207}Pb/^{206}Pb$ date of 2059.05 ± 0.50 (MSWD = 0.34), which is slightly older than, but within error of the weighted mean $^{207}Pb/^{206}Pb$ zircon date.

Discussion

Single-step digestions

The two sets of single-step digestion experiments demonstrate the complexity of the Neoproterozoic Gannakouriep baddeleyite. Data from un-annealed and annealed grains yield upper intercept dates of 793.2 ± 1.0 and 787.4 + 1.3/−1.2 Ma, respectively, and both data sets had higher MSWD than would be expected for simple two component mixing (Fig. 3). The significant offset in the upper intercept dates could reflect either heterogeneities in the analysed baddeleyite population, such as variable proportions of over-grown zircon, or an unpredicted result of the annealing process. However, several lines of evidence strongly favour the former explanation: (1) the inferred slow diffusion rates for U and Pb in baddeleyite and the short duration (60 h) of the annealing; (2) the upper intercept date of the HCl steps from the discrete digestion experiments on annealed baddeleyite is within error of the un-annealed analyses from the single-step digestions and (3) analyses of annealed Phalaborwa baddeleyite yield $^{207}Pb/^{206}Pb$ dates that are within error of previously published results on un-annealed grains (Heaman 2009) and the results presented here on un-annealed baddeleyite from the PhalaHMS separate (Fig. 8). In general, the high MSWDs for the un-annealed (MSWD = 3.1), annealed (MSWD = 3.0) and combined (MSWD = 6.7) data sets, the scatter of the data towards lower $^{207}Pb/^{235}U$ and the offset upper intercept dates of the two data sets are all

Table 3 Discrete HCl–HF digestion experiments on 05-48A

	Digestion conditions[a]			Composition					Isotopic ratios					
	Reag.	T (°C)	t (h)	Pb*/Pbc[b]	Pb* (pg)[c]	Pbc (pg)[c]	U (pg)[c]	Th/U[d]	$^{206}Pb/^{204}Pb$[e]	$^{208}Pb/^{206}Pb$[f]	$^{206}Pb^*/^{238}U$[f]	2σ (% err)	$^{207}Pb^*/^{235}U$[f]	2σ (% err)
05-48A, first-stage 6.2 N HCl digestion[j]														
b25	HCl	210	48	27.1	383.22	14.13	3353.43	0.050	1823	0.01534	0.123021	0.060	1.1105	0.138
b26	HCl	210	48	2.1	185.98	88.70	1695.59	0.070	157	0.02157	0.117372	0.110	1.0626	1.168
B27	HCl	210	48	8.9	708.74	79.28	6012.96	0.025	617	0.00775	0.127790	0.079	1.1541	0.301
b28	HCl	210	48	22.0	318.46	14.48	2667.13	0.026	1491	0.00788	0.129432	0.061	1.1694	0.160
b29	HCl	210	48	4.0	110.12	27.57	999.87	0.057	283	0.01748	0.118329	0.077	1.0679	0.618
b32	HCl	210	48	10.3	39.58	3.84	331.73	0.016	712	0.00487	0.129676	0.073	1.1738	0.296
b33	HCl	210	48	4.5	89.33	19.80	750.75	0.031	320	0.00950	0.128785	0.157	1.1637	0.639
b34	HCl	210	48	43.3	200.45	4.63	1675.96	0.016	2928	0.00495	0.129997	0.052	1.1760	0.119
b35	HCl	210	48	16.1	246.28	15.33	2062.42	0.014	1098	0.00433	0.129880	0.055	1.1733	0.193
b37	HCl	210	48	2.2	25.87	11.69	217.79	0.025	166	0.00775	0.128758	0.112	1.1701	1.160
b45	HCl	210	48	31.3	93.43	2.98	824.12	0.014	2127	0.00439	0.123304	0.073	1.1135	0.192
b46	HCl	210	48	5.5	79.09	14.43	670.80	0.027	385	0.00830	0.127754	0.123	1.1552	0.533
b49	HCl	210	48	4.2	46.34	11.07	498.70	0.232	283	0.07219	0.094991	0.083	0.8586	0.636
b51	HCl	210	48	15.3	106.83	6.98	900.02	0.014	1048	0.00420	0.129120	0.076	1.1663	0.266
05-48A, second-stage 29 N HF leach[j]														
b30 lch	HF	120	7	0.9	1.61	1.80	16.55	0.603	70	0.18737	0.091058	3.031	0.6394	9.364
b35 lch	HF	120	7	1.7	3.43	2.06	57.49	0.451	118	0.14175	0.057734	0.413	0.4753	3.018
b45 lch	HF	140	8	2.6	4.29	1.66	49.42	0.035	193	0.01082	0.094342	0.273	0.7882	1.701
b47 lch	HF	140	8	2.1	2.17	1.06	33.79	0.349	145	0.10953	0.063802	0.400	0.5503	3.225
b48 lch	HF	140	8	1.4	1.27	0.94	11.94	0.734	94	0.22782	0.095826	0.754	0.8438	5.801
b50 lch	HF	140	8	2.5	2.53	1.02	22.32	0.067	182	0.02070	0.121106	0.628	1.1026	2.710
05-48A, final 29 N HF digestion[j]														
b25	HF	210	48	90.9	215.45	2.37	1849.46	0.770	5041	0.23833	0.104014	0.056	0.8997	0.119
b26	HF	210	48	77.8	377.92	4.86	2639.88	1.328	3809	0.41013	0.112728	0.057	0.9914	0.113
b27	HF	210	48	73.1	220.05	3.01	2080.61	0.630	4193	0.19548	0.097565	0.062	0.8625	0.128
b28	HF	210	48	35.9	21.37	0.60	171.82	0.622	2082	0.19206	0.115089	0.099	1.0118	0.284
b29	HF	210	48	19.1	108.87	5.69	784.99	0.984	1022	0.30321	0.117779	0.063	1.0428	0.216
b30[k]	HF	210	48	6.4	19.70	3.10	144.59	0.869	361	0.26776	0.118848	0.130	1.0495	0.598
b31	HF	210	48	7.5	4.60	0.61	41.65	0.330	485	0.10182	0.110291	0.533	0.9642	1.221
b32	HF	210	48	6.6	8.56	1.29	79.40	0.073	457	0.02244	0.115433	0.194	1.0149	0.900
b33	HF	210	48	39.2	34.40	0.88	247.69	1.142	2014	0.35262	0.113783	0.082	1.0104	0.306
b34	HF	210	48	27.6	36.01	1.30	257.52	1.043	1453	0.32153	0.117171	0.078	1.0392	0.286
b35[k]	HF	210	48	37.0	48.23	1.30	413.09	0.629	2138	0.19437	0.107863	0.066	0.9410	0.243

Table 3 continued

	Digestion conditions[a]			Composition					Isotopic ratios					
	Reag.	T (°C)	t (h)	Pb*/Pbc[b]	Pb* (pg)[c]	Pbc (pg)[c]	U (pg)[c]	Th/U[d]	$^{206}Pb/^{204}Pb$[e]	$^{208}Pb/^{206}Pb$[f]	$^{206}Pb^*/^{238}U$[f]	2σ (% err)	$^{207}Pb^*/^{235}U$[f]	2σ (% err)
b45[k]	HF	210	48	18.3	9.41	0.51	105.96	0.041	1253	0.01282	0.096408	0.101	0.7856	0.461
b46[k]	HF	210	48	23.6	22.16	0.94	174.27	0.484	1420	0.14890	0.121811	0.117	1.0790	0.427
b47[k]	HF	210	48	18.8	17.11	0.91	137.11	0.625	1096	0.19272	0.115391	0.128	1.0114	0.540
b48[k]	HF	210	48	14.6	10.19	0.70	83.56	0.737	836	0.22770	0.109741	0.131	0.9524	0.664
b49[k]	HF	210	48	37.0	117.18	3.17	885.52	1.030	1939	0.31819	0.111171	0.056	0.9800	0.154
b50[k]	HF	210	48	12.6	7.52	0.60	64.21	0.190	827	0.05841	0.121314	0.176	1.0792	0.660
b51[k]	HF	210	48	10.9	6.32	0.58	55.56	0.062	748	0.01900	0.122116	0.164	1.0922	0.697

	Isotopic ratios			Dates						
	$^{207}Pb^*/^{206}Pb^*$[f]	2σ (% err)	corr.[g] coef.	$^{206}Pb/^{238}U$[h]	2σ (abs)	$^{207}Pb/^{235}U$[h]	2σ (abs)	$^{207}Pb/^{206}Pb$[h]	2σ (abs)	% disc[i]
05-48A, first-stage 6.2 N HCl digestion[j]										
b25	0.065468	0.108	0.655	747.93	0.42	758.40	0.74	789.41	2.28	5.26
b26	0.065659	1.117	0.498	715.42	0.74	735.09	6.11	795.52	23.43	10.07
b27	0.065498	0.271	0.492	775.25	0.58	779.15	1.64	790.37	5.70	1.91
b28	0.065527	0.128	0.658	784.63	0.45	786.37	0.88	791.31	2.70	0.85
b29	0.065451	0.584	0.493	720.94	0.52	737.69	3.24	788.88	12.25	8.61
b32	0.065648	0.264	0.525	786.02	0.54	788.41	1.62	795.17	5.55	1.15
b33	0.065536	0.589	0.429	780.93	1.15	783.70	3.49	791.59	12.36	1.35
b34	0.065611	0.081	0.806	787.85	0.39	789.46	0.65	794.00	1.73	0.77
b35	0.065520	0.161	0.662	787.18	0.41	788.20	1.06	791.08	3.39	0.49
b37	0.065911	1.107	0.508	780.78	0.83	786.70	6.35	803.55	23.18	2.83
b45	0.065495	0.149	0.699	749.55	0.52	759.86	1.03	790.29	3.15	5.15
b46	0.065583	0.483	0.502	775.04	0.89	779.71	2.90	793.11	10.15	2.28
b49	0.065557	0.599	0.491	584.99	0.47	629.37	2.98	792.25	12.57	26.16
b51	0.065512	0.227	0.612	782.85	0.56	784.92	1.45	790.82	4.77	1.01
05-48A, second-stage 29 N HF leach[j]										
b30 lch	0.050925	8.380	0.469	561.79	16.31	501.91	37.08	237.42	193.29	−136.63
b35 lch	0.059703	2.832	0.505	361.83	1.45	394.81	9.87	592.82	61.39	38.96
b45 lch	0.060591	1.588	0.479	581.16	1.52	590.13	7.61	624.75	34.25	6.98
b47 lch	0.062552	3.038	0.517	398.70	1.55	445.17	11.62	693.05	64.77	42.47
b48 lch	0.063860	5.440	0.528	589.90	4.25	621.21	26.95	736.99	115.15	19.96
b50 lch	0.066029	2.468	0.484	736.92	4.38	754.59	14.43	807.30	51.65	8.72
05-48A, final 29 N HF digestion[j]										
b25	0.062734	0.080	0.807	637.89	0.34	651.57	0.57	699.23	1.73	8.77

Table 3 continued

	Isotopic ratios			Dates						
	$^{207}Pb^*/^{206}Pb^*$[f]	2σ (% err)	corr.[g] coef.	$^{206}Pb/^{238}U$[h]	2σ (abs)	$^{207}Pb/^{235}U$[h]	2σ (abs)	$^{207}Pb/^{206}Pb$[h]	2σ (abs)	% disc[i]
b26	0.063786	0.072	0.825	688.57	0.37	699.45	0.57	734.55	1.56	6.26
b27	0.064116	0.092	0.727	600.12	0.35	631.49	0.60	745.47	1.97	19.50
b28	0.063759	0.248	0.513	702.24	0.66	709.77	1.45	733.66	5.27	4.28
b29	0.064216	0.185	0.597	717.77	0.43	725.33	1.12	748.77	3.92	4.14
b30[k]	0.064044	0.543	0.509	723.93	0.89	728.62	3.11	743.07	11.49	2.58
b31	0.063403	1.032	0.546	674.44	3.41	685.45	6.09	721.78	21.89	6.56
b32	0.063764	0.831	0.449	704.22	1.29	711.33	4.60	733.82	17.61	4.03
b33	0.064402	0.271	0.537	694.68	0.54	709.06	1.56	754.85	5.73	7.97
b34	0.064323	0.246	0.606	714.26	0.53	723.51	1.48	752.28	5.20	5.05
b35[k]	0.063272	0.200	0.727	660.32	0.41	673.40	1.20	717.37	4.25	7.95
b45[k]	0.059100	0.422	0.473	593.32	0.57	588.68	2.06	570.79	9.19	−3.95
b46[k]	0.064245	0.377	0.543	740.98	0.82	743.16	2.25	749.72	7.96	1.17
b47[k]	0.063573	0.495	0.454	703.98	0.85	709.60	2.76	727.45	10.50	3.23
b48[k]	0.062944	0.610	0.499	671.25	0.83	679.36	3.29	706.35	12.97	4.97
b49[k]	0.063935	0.122	0.694	679.55	0.36	693.61	0.78	739.47	2.59	8.10
b50[k]	0.064517	0.596	0.479	738.12	1.23	743.22	3.48	758.61	12.57	2.70
b51[k]	0.064868	0.637	0.467	742.73	1.15	749.58	3.69	770.06	13.41	3.55

[a] All digestions carried out in Parr acid digestion vessels

[b] Ratio of radiogenic to common Pb

[c] Total radiogenic Pb, common Pb, and U (picograms)

[d] Th/U ratio calculated from $^{208}Pb/^{206}Pb$ and the $^{206}Pb/^{238}U$ date of the sample

[e] Fractionation and spike corrected $^{206}Pb/^{204}Pb$

[f] Fractionation, spike, and common Pb corrected radiogenic isotope ratios. Initial Pbc was corrected using $^{206}Pb/^{204}Pb = 17.89 \pm 0.10$, $^{207}Pb/^{204}Pb = 15.59 \pm 0.10$, and $^{208}Pb/^{204}Pb = 37.76 \pm 0.20$ (2σ absolute errors). Laboratory Pb blanks were ~0.72 pg and were corrected using $^{206}Pb/^{204}Pb = 18.24 \pm 0.42$, $^{207}Pb/^{204}Pb = 15.34 \pm 0.31$, and $^{208}Pb/^{204}Pb = 37.35 \pm 0.99$ (2σ absolute errors)

[g] Correlation coefficient of radiogenic $^{207}Pb^*/^{235}U$ and $^{206}Pb^*/^{238}U$

[h] Dates (Ma) calculated using $^{238}U/^{235}U = 137.88$, and decay constants of $^{238}U = 1.5513 \times 10^{-10}$ year^{-1} and $^{235}U = 9.8485 \times 10^{-10}$ year^{-1}. Uncertainties are 2σ absolute errors

[i] % discordance = 100 − (100 × ($^{206}Pb/^{238}U$ date)/($^{207}Pb/^{206}Pb$ date))

[j] Individual grains were subjected to two- or three-step digestions as described in the text. Missing HF or HCl data indicate that the digestion step did not generate usable data

[k] Zircon rims subjected to a 120–140°C leach in 29 N HF for 6–8 h prior to final digestion

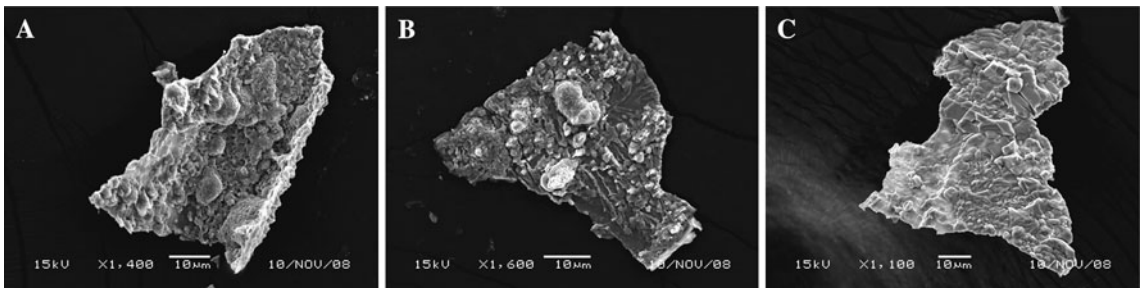


Fig. 6 Fragments of a zircon rim left after baddeleyite was dissolve in 6.2 N HCl (48 h-220°C). **a–b** Inside surface of rim fragments. **c** Outside surface of the rim

consistent with a combination of post-crystallization Pb-loss and mixing between the baddeleyite and zircon components identified in the discrete HCl–HF digestion experiments.

The paired analyses of HNO_3-rinsed and H_2O-rinsed grain fragments suggest that the low-temperature HNO_3 fluxing prior to digestion did not have a significant impact on the U–Pb dates. The HNO_3- versus H_2O-rinsed fragments from individual grains did not have systematic offsets (Fig. 4), and the variability in the $^{206}Pb/^{238}U$ and $^{207}Pb/^{235}U$ dates between pieces of a single grain may simply reflect different proportions of baddeleyite and zircon in the analysed fragments. This interpretation is further supported by the Phalaborwa hot plate experiment. The data from this experiment indicate that some radiogenic Pb was removed during the hot plate leach, however, it was a tiny fraction (0.009%) of the total Pb* in the grain and may have come from either the baddeleyite or other actinide-rich mineral inclusions within the grain. Taken together, these data suggest that short rinses in dilute acid at low temperatures, which are commonly used to clean grains prior to digestion, are unlikely to have a significant impact on the U–Pb systematics of most baddeleyite.

Multi-step baddeleyite chemical abrasion

The power of the chemical abrasion approach in zircon U–Pb geochronology was originally demonstrated in multi-step digestion experiments on large zircon populations (Mattinson 2005). In these analyses, initial low-temperature digestion steps remove damaged zircon and yield variably discordant data. After initial digestion of highly soluble domains, the $^{206}Pb/^{238}U$ and $^{207}Pb/^{235}U$ of the analyses typically increase sequentially until the sample reaches an age plateau, where successive digestion steps are concordant and overlap within error (e.g., Mattinson 2005; Rioux et al. 2007). The reproducible high-temperature digestion steps are interpreted to reflect the crystallization age of the sample.

The multi-step baddeleyite experiments generated more complex results than typical zircon analyses. In all three experiments, the early digestion steps removed material with lower $^{206}Pb/^{238}U$ and $^{207}Pb/^{235}U$ than the corresponding solid, and each successive digestion moved progressively closer to concordia. However, the final high-temperature digestion steps prior to complete digestion of the residue were either reversely discordant (HCl experiments) or had lower $^{206}Pb/^{238}U$ and $^{207}Pb/^{235}U$ than the preceding analyses (HNO_3 experiment) and do not record the consistent age plateau seen in multi-step zircon data.

The reversely discordant data from the HCl digestions require either excess Pb* or U loss in the high-temperature steps. Reversely discordant analyses are common for early digestion steps of un-annealed zircon and are interpreted to reflect leaching of excess Pb* from damaged high-U zones within the analysed grains (Chen et al. 2002; Mattinson 2005). In addition, Mattinson et al. (1996) argued that true reverse discordance in zircon can be generated by fine-scale U zoning, where Pb* can be transported from high-U to adjacent low-U zones by either thermally activated volume diffusion or implantation. In general, Pb redistribution in baddeleyite and zircon is a more likely mechanism than U-loss for generating reverse discordance, because U is structurally bound within the crystal lattice of these minerals, while the Pb* atoms occur at the end of radiation damage tracks. As a result, we interpret the reversely discordant data from the multi-step baddeleyite analyses to reflect leaching of Pb* from the residual solids.

The evidence for decoupling of the U–Pb system limits the application of multi-step chemical abrasion to baddeleyite. In the multi-step experiments, the initial low-temperature digestion steps had lower $^{206}Pb/^{238}U$ and $^{207}Pb/^{235}U$ than the bulk grain, suggesting that these steps did preferentially dissolved damaged baddeleyite that had lost Pb. However, the evidence for unsupported Pb* in the higher temperature digestion steps indicates that the U and Pb from each step may include a mixture of U and Pb from dissolved baddeleyite and leached Pb* from more

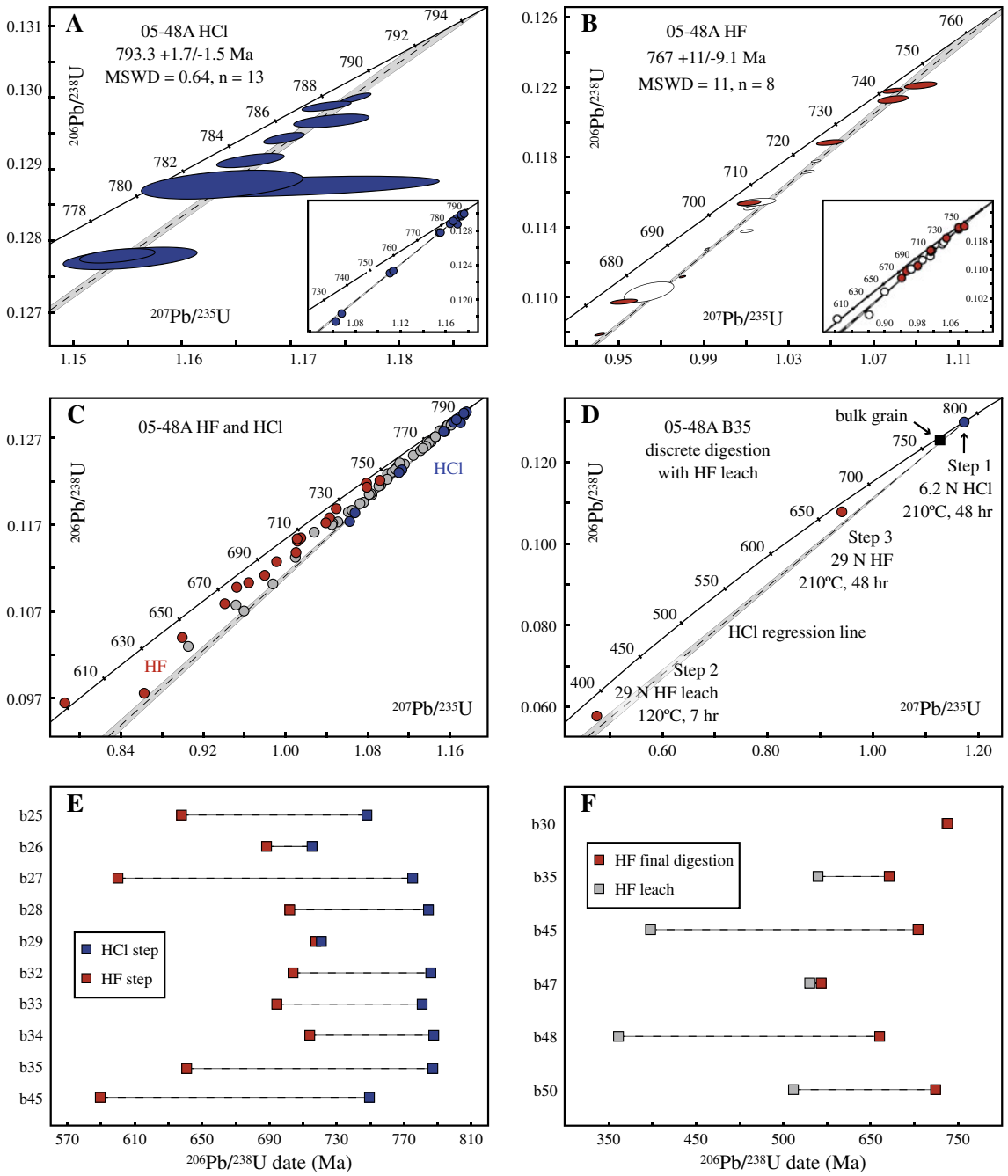


Fig. 7 U–Pb concordia diagrams of the discrete HCl–HF digestion results. *Inset* plots show the full data sets. Open data points were excluded from the regressions. *Plot symbol colours* correspond to the reagent used in the digestion (*blue*, HCl; *red*, HF). **a** Data from initial 210°C-48 h 6.2 N HCl digestion of baddeleyite. A single highly discordant grain (b49) was excluded from the plot and regression. **b** Data from final 210°C-48 h HF digestions of the zircon rims and intergrowths. **c** Comparison of the HF and HCl results. The HCl regression line from **a** is shown for reference. The *grey symbols* are data from the single-step digestions for this sample (Fig. 3; Table 1). **d** An example of data from a three-step discrete digestion analysis. The bulk grain composition is calculated by mass balance. The HCl regression line from **a** is shown for reference. **e** Comparison of the $^{206}Pb/^{238}U$ dates from the HCl and HF digestion steps. For grains where the zircon was treated with an intermediate HF leach step, the two HF steps were recombined by mass balance. Samples that are missing leach or HF digestion data were excluded from the plot. **f** Comparison of the $^{206}Pb/^{238}U$ dates for the HF leach and final HF digestions steps from the three-step digestions. Ages along concordia are in Ma. All error ellipses are 2-sigma. Concordia plots and regressions were generated in U–Pb Redux (Bowring et al. 2008; McLean et al. 2008)

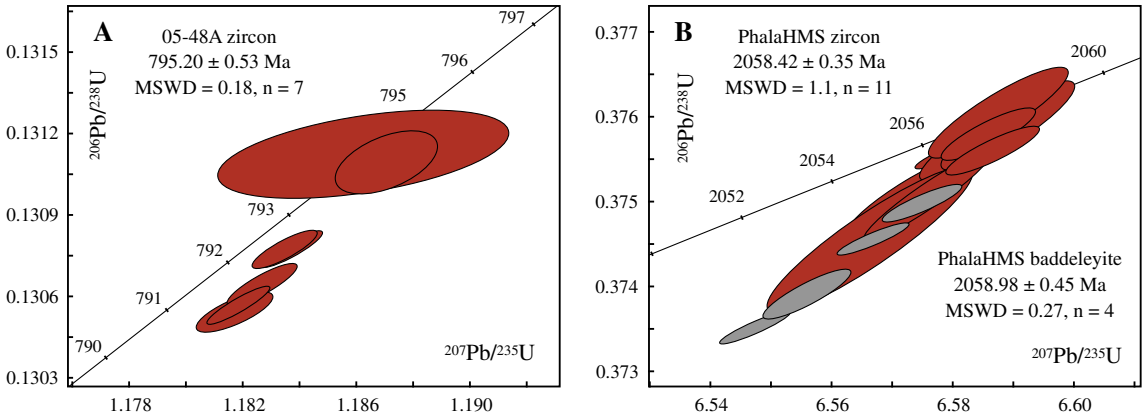


Fig. 8 U–Pb concordia diagrams of igneous zircon (*red*) and baddeleyite (*grey*) data from samples 05-48A and PhalaHMS. The igneous zircon from sample 05-48A were found in the same mineral separate as the 05-48A baddeleyite. The PhalaHMS zircon and baddeleyite were separated from a heavy mineral separate from the Phalaborwa mine, within the same carbonatite complex as the PhalaSG sample. Ages along concordia are in Ma. All error ellipses are 2-sigma. Concordia plots and regressions were generated in U–Pb Redux (Bowring et al. 2008; McLean et al. 2008)

refractory material. As a result, the measured U/Pb ratios from individual digestion steps do not necessarily reflect the age of the dissolved material. It may be possible in some cases to optimize digestion procedures to maximize the removal of discordant baddeleyite while minimizing the leaching of unsupported Pb*. However, the complexity of the Phalaborwa data indicates that the multi-step chemical abrasion method, in its current form, is not generally applicable to baddeleyite geochronology.

Discrete digestion experiments

In contrast to the multi-step chemical abrasion experiments, the HCl–HF discrete digestion experiments were highly successful and provide a new approach for isolating zircon and baddeleyite in composite grains. Previous studies of intergrown baddeleyite and zircon have relied on identifying zircon-free grains to constrain the baddeleyite crystallization age and isolating rim fragments to determine the zircon age (Davidson and van Breemen 1988). However, for intimately intergrown baddeleyite and zircon or small grain sizes, physical separation can be difficult or impossible (Davidson and van Breemen 1988; Heaman and LeCheminant 1993). The discrete HCl and HF digestions for the Gannakouriep grains effectively chemically separated the baddeleyite and zircon components.

An important consideration in the discrete digestion analyses is the potential impact of the initial HCl (baddeleyite) digestion step on the zircon rims. Previous studies have demonstrated that for the annealing conditions used in this study, metamict zircon transforms to amorphous silica and a mixture of monoclinic and metastable tetragonal ZrO_2 (e.g. Nasdala et al. 2002). As a result, any metamict zones within the zircon rims and intergrowths could be susceptible to dissolution during the initial HCl digestion step. For sample 05-48A, several factors suggest that this was not an important effect: (1) post-HCl SEM images document relatively coherent zircon rims (Fig. 6); (2) the HCl and HF data plot along distinct trends and (3) the HCl data define a statistically significant discordia with an upper intercept date (793.3 = 1.7/−1.5 Ma; Fig. 7) that is within error of the weighted mean $^{207}Pb/^{206}Pb$ date of igneous zircon from the same sample (795.2 ± 0.5 Ma; Fig. 8). Instead, the data suggest that the HCl digestion steps for this sample were dominated by U and Pb from igneous baddeleyite. The slight offset between the baddeleyite and igneous zircon dates may reflect incorporation of some Pb and U from metamic zircon dissolved during the HCl digestions, but can also be explained by statistical variability or a true offset in the zircon and baddeleyite ages. Dissolution of metamict zircon may have a larger impact on baddeleyite dates for other samples with highly damaged zircon rims and should be taken into consideration when interpreting data from such samples.

For non-metamict zircon, the annealing process repairs most of the radiation damage in the crystal structure and prevents leaching of Pb*. Multi-step chemical abrasion experiments routinely include high-temperature HCl fluxes in Parr acid digestion vessels to clean the zircons between digestion steps and indicate that the initial HCl digestion step in the discrete digestion analyses should not impact the U–Pb systematics of non-metamic zircon (Mattinson 2005).

The three-step discrete digestion experiments show the potential for further refining the zircon U–Pb dates for composite grains. These experiments included an initial HCl digestion step, a low-temperature HF leach step and then a final high-temperature HF digestion step. The two-

Table 4 Zircon (z) and baddeleyite (b) data from 05-48A and PhalaHMS

	Leach[a]		Composition					Isotopic ratios					
	T (°C)	*t* (h)	Pb*/Pbc[b]	Pb* (pg)[c]	Pbc(pg)[c]	U (pg)[c]	Th/U[d]	$^{206}Pb/^{204}Pb$[e]	$^{208}Pb/^{206}Pb$[f]	$^{206}Pb^*/^{238}U$[f]	2σ (% err)	$^{207}Pb^*/^{235}U$[f]	2σ (% err)
05-48A													
z1	180	12	508.9	206.17	0.41	1526.19	0.440	30646	0.13512	0.130774	0.052	1.1837	0.093
z2	180	12	38.7	24.03	0.62	99.28	3.492	1322	1.07200	0.131124	0.124	1.1862	0.432
z4	180	12	187.9	156.59	0.83	995.95	1.084	9715	0.33274	0.130542	0.057	1.1817	0.114
z7	180	12	583.7	226.78	0.39	1530.07	0.821	32002	0.25214	0.130643	0.060	1.1827	0.104
z8	180	12	738.9	183.70	0.25	1218.20	0.891	39856	0.27358	0.130773	0.054	1.1835	0.097
z9	180	12	340.1	78.06	0.23	523.53	0.832	18609	0.25544	0.131094	0.089	1.1871	0.152
z10	180	12	1050.0	271.35	0.26	1857.92	0.761	58373	0.23382	0.130567	0.054	1.1818	0.094
PhalaHMS													
z1	180	12	373.5	82.02	0.22	182.53	0.923	19398	0.26425	0.375079	0.115	6.5721	0.140
z2	180	12	1451.6	368.32	0.25	707.31	1.694	64959	0.48503	0.374743	0.088	6.5674	0.116
z11	210	12	790.9	374.58	0.47	749.44	1.458	36966	0.41727	0.375585	0.053	6.5800	0.095
z12	210	12	198.6	56.75	0.29	123.71	1.015	10131	0.29055	0.375726	0.128	6.5846	0.153
z13	210	12	183.6	63.91	0.35	133.80	1.216	9001	0.34801	0.375896	0.140	6.5891	0.167
z14	210	12	161.9	48.65	0.30	60.15	4.762	4698	1.36299	0.376048	0.142	6.5875	0.175
z15	210	12	69.2	39.93	0.58	93.31	0.696	3788	0.19919	0.374821	0.104	6.5713	0.149
z16	210	12	78.0	30.21	0.39	57.72	1.725	3490	0.49391	0.374553	0.229	6.5663	0.258
z17	210	12	471.7	270.43	0.57	483.19	2.107	19672	0.60328	0.375061	0.139	6.5758	0.159
z18	210	12	163.0	119.58	0.73	277.41	0.718	8853	0.20544	0.375795	0.082	6.5859	0.118
z19	210	12	144.3	89.63	0.62	208.08	0.717	7841	0.20515	0.375601	0.076	6.5866	0.117
b6	–	–	203.6	102.35	0.50	280.41	0.029	12714	0.00843	0.373911	0.079	6.5559	0.111
b7	–	–	391.2	166.53	0.43	456.69	0.014	24874	0.00408	0.374982	0.059	6.5749	0.100
b9	–	–	181.6	380.66	2.10	1042.83	0.023	10705	0.00653	0.374560	0.051	6.5669	0.090
b10	–	–	677.2	294.70	0.44	808.31	0.029	42793	0.00834	0.373516	0.052	6.5474	0.089

	Isotopic ratios			Dates						
	$^{207}Pb^*/^{206}Pb^*$[f]	2σ (% err)	corr.[g] coef.	$^{206}Pb/^{238}U$[h]	2σ (abs)	$^{207}Pb/^{235}U$[h]	2σ (abs)	$^{207}Pb/^{206}Pb$[h]	2σ (abs)	% disc[i]
05-48A										
z1	0.065648	0.051	0.885	792.28	0.39	793.04	0.51	795.18	1.11	0.36
z2	0.065613	0.388	0.479	794.27	0.93	794.22	2.38	794.06	8.14	−0.03
z4	0.065654	0.079	0.754	790.96	0.42	792.11	0.63	795.36	1.69	0.55
z7	0.065657	0.059	0.862	791.54	0.44	792.56	0.57	795.45	1.28	0.49
z8	0.065636	0.057	0.848	792.27	0.40	792.93	0.53	794.78	1.23	0.32
z9	0.065673	0.131	0.504	794.10	0.66	794.60	0.84	795.98	2.76	0.24

Table 4 continued

	Isotopic ratios			Dates						
	$^{207}Pb^*/^{206}Pb^*$ [f]	2σ (% err)	corr. [g] coef.	$^{206}Pb/^{238}U$ [h]	2σ (abs)	$^{207}Pb/^{235}U$ [h]	2σ (abs)	$^{207}Pb/^{206}Pb$ [h]	2σ (abs)	% disc [i]
z10	0.065649	0.048	0.911	791.10	0.40	792.17	0.52	795.20	1.05	0.52
PhalaHMS										
z1	0.127081	0.053	0.925	2053.25	2.02	2055.61	1.23	2057.98	0.97	0.23
z2	0.127104	0.052	0.900	2051.68	1.54	2054.98	1.02	2058.30	0.94	0.32
z11	0.127063	0.049	0.912	2055.63	0.93	2056.68	0.84	2057.73	0.91	0.10
z12	0.127103	0.070	0.884	2056.29	2.25	2057.28	1.35	2058.28	1.27	0.10
z13	0.127133	0.080	0.876	2057.08	2.47	2057.89	1.48	2058.70	1.43	0.08
z14	0.127051	0.082	0.883	2057.80	2.51	2057.68	1.54	2057.56	1.46	−0.01
z15	0.127152	0.086	0.819	2052.05	1.82	2055.50	1.31	2058.96	1.54	0.34
z16	0.127148	0.106	0.910	2050.79	4.03	2054.84	2.28	2058.90	1.90	0.39
z17	0.127158	0.052	0.943	2053.17	2.44	2056.10	1.40	2059.05	0.95	0.29
z18	0.127105	0.063	0.855	2056.61	1.45	2057.46	1.04	2058.31	1.13	0.08
z19	0.127184	0.060	0.879	2055.70	1.35	2057.55	1.03	2059.41	1.09	0.18
b6	0.127165	0.055	0.875	2047.77	1.38	2053.44	0.98	2059.14	1.00	0.55
b7	0.127169	0.053	0.881	2052.80	1.04	2055.99	0.88	2059.20	0.98	0.31
b9	0.127155	0.047	0.909	2050.82	0.89	2054.91	0.80	2059.01	0.87	0.40
b10	0.127132	0.044	0.920	2045.92	0.91	2052.29	0.79	2058.69	0.82	0.62

[a] Zircon leach temperatures and duration. All zircon were annealed at 900°C for 60 h prior to digestion

[b] Ratio of radiogenic to common Pb

[c] Total radiogenic Pb, common Pb, and U (picograms)

[d] Th/U ratio calculated from $^{208}Pb/^{206}Pb$ and the $^{206}Pb/^{238}U$ date of the sample

[e] Fractionation and spike corrected $^{206}Pb/^{204}Pb$

[f] Fractionation, spike, and common Pb corrected radiogenic isotope ratios. For the baddeleyite analyses, initial Pbc was corrected using $^{206}Pb/^{204}Pb = 15.04 \pm 0.30$, $^{207}Pb/^{204}Pb = 15.16 \pm 0.30$ and $^{208}Pb/^{204}Pb = 34.67 \pm 0.69$ (2σ absolute errors). Laboratory Pb blanks ranged from 0.22 to 0.83 pg and were corrected using $^{206}Pb/^{204}Pb = 18.24 \pm 0.42$, $^{207}Pb/^{204}Pb = 15.34 \pm 0.31$ and $^{208}Pb/^{204}Pb = 37.35 \pm 0.99$ (2σ absolute errors)

[g] Correlation coefficient of radiogenic $^{207}Pb^*/^{235}U$ and $^{206}Pb^*/^{238}U$

[h] Dates (Ma) calculated using $^{238}U/^{235}U = 137.88$, and decay constants of $^{238}U = 1.5513 \times 10^{-10}$ year^{-1} and $^{235}U = 9.8485 \times 10^{-10}$ year^{-1}. Uncertainties are 2σ absolute errors

[i] % discordance = 100 − (100 × ($^{206}Pb/^{238}U$ date)/($^{207}Pb/^{206}Pb$ date))

step HF digestion of the zircon rims replicate single grain chemical abrasion procedures that are now routinely used in ID-TIMS zircon dating. Analysed HF leachates had lower $^{206}Pb/^{238}U$ and $^{207}Pb/^{235}U$ than the final zircon digestions, indicating that the intermediate leach step removed damage zircon and shifted the final zircon residue towards concordia (Fig. 7d, f; Table 3). For the analysed sample, the thin zircon rims were relatively soluble in HF, and we were not able to isolate undisturbed concordant zircon. However, the three-step procedure shows promise for other samples.

The results from the discrete digestion and multi-step chemical abrasion experiments suggest that baddeleyite can be affected by significant post-crystallization Pb-loss. In the multi-step chemical abrasion experiments, the low-temperature digestion steps routinely yielded highly discordant data, consistent with dissolution of damage zones that have lost Pb*. In the discrete digestion experiments, the HCl digestion steps, which preferentially dissolved the baddeleyite in the composite grains, range from 0.45 to 9.75% discordance. The spread of the data along a statistically significant discordia is consistent with either a recent Pb-loss event or mixing between younger baddeleyite rims and older baddeleyite cores. The latter explanation is unlikely due to the absence of visible zoning within the baddeleyite in BSE images and the evidence for the growth of zircon rims, instead of baddeleyite rims, during later metamorphism or alteration (Fig. 1). In addition, the distinct trends of the HCl (baddeleyite) and HF (zircon) analyses in the discrete digestions experiments make it unlikely that the observed discordance in the HCl digestion steps reflects mixing between igneous baddeleyite and a metamic zircon component (discussed above). Instead, the discordant analyses suggest that baddeleyite, like zircon, can be affected by fast pathway diffusion of Pb leading to variable degree of post-crystallization Pb-loss.

Overall, the discrete digestion experiments demonstrate that the combined HCl and HF digestions can isolate both igneous baddeleyite and igneous or metamorphic zircon dates from composite grains. For the Gannakouriep baddeleyite, the variable upper intercept dates and excessive scatter in the single-step analyses demonstrate how secondary zircon in composite grains can lead to inaccurate baddeleyite crystallization ages. In the discrete digestion experiments, data from the HCl (baddeleyite) digestion steps define a statistically significant discordia and we interpret the upper intercept date ($793.3 = 1.7/-1.5$ Ma; Fig. 7) to be the best estimate of the igneous crystallization age of the baddeleyite. The data from the HF (zircon) digestion steps have excessive scatter and systematically younger $^{207}Pb/^{206}Pb$ dates, suggesting multiple periods of Pb-loss or zircon growth. The complexity of these data may reflect the extremely small grain sizes and high surface area of the fine-grained zircon over-growths, which could facilitate Pb-loss. Secondary zircon inter- and over-growths of baddeleyite have been observed in samples affected by low-temperature alteration to granulite facies metamorphism (Davidson and van Breemen 1988; Heaman and Grotzinger 1992; Heaman and LeCheminant 1993), and may be more common than currently recognized. The HCl–HF discrete digestion method provides a means for generating more precise and accurate ages from these samples.

Conclusion

Baddeleyite digestion experiments were carried out to understand and reduce the impact of Pb-loss and secondary zircon growth on baddeleyite geochronology. Multi-step HCl and HNO_3 chemical abrasion analyses of Phalaborwa baddeleyite yielded complex results: Individual digestion steps were both normally and reversely discordant suggesting that U and Pb were decoupled in the digestion. The data indicate that multi-step chemical abrasion in its current form cannot be routinely applied to baddeleyite geochronology. In contrast, the newly developed HCl–HF discrete digestion method proved successful at isolating the zircon and baddeleyite components of composite grains. Single-step digestion data from sample 05-48A highlight the difficulty of obtaining precise and accurate data from baddeleyites that are over-grown by zircon rims: statistical regressions of annealed and un-annealed single-step data generated higher than expected MSWD and distinct upper intercept dates. In the discrete digestion method, initial HCl digestion steps completely dissolved the baddeleyite grains with minimal impact to the zircon rims and intergrowths. The zircon was then dissolved in a single- or two-step HF digestion to determine the date of the igneous or metamorphic over-growths. For sample 05-48A, the baddeleyite (HCl) data define a statistically significant discordia and we interpret the upper intercept date to be the best estimate of the baddeleyite crystallization age. In contrast, the zircon rims and inter-growths from the baddeleyites had systematically younger $^{207}Pb/^{206}Pb$ dates consistent with either later zircon growth or non-zero-age Pb-loss. Zircon inter- and over-growths of baddeleyite can form under a wide range of metamorphic conditions, and the new discrete HCl–HF digestion technique has the potential to provide meaningful ages for a range of complex samples.

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